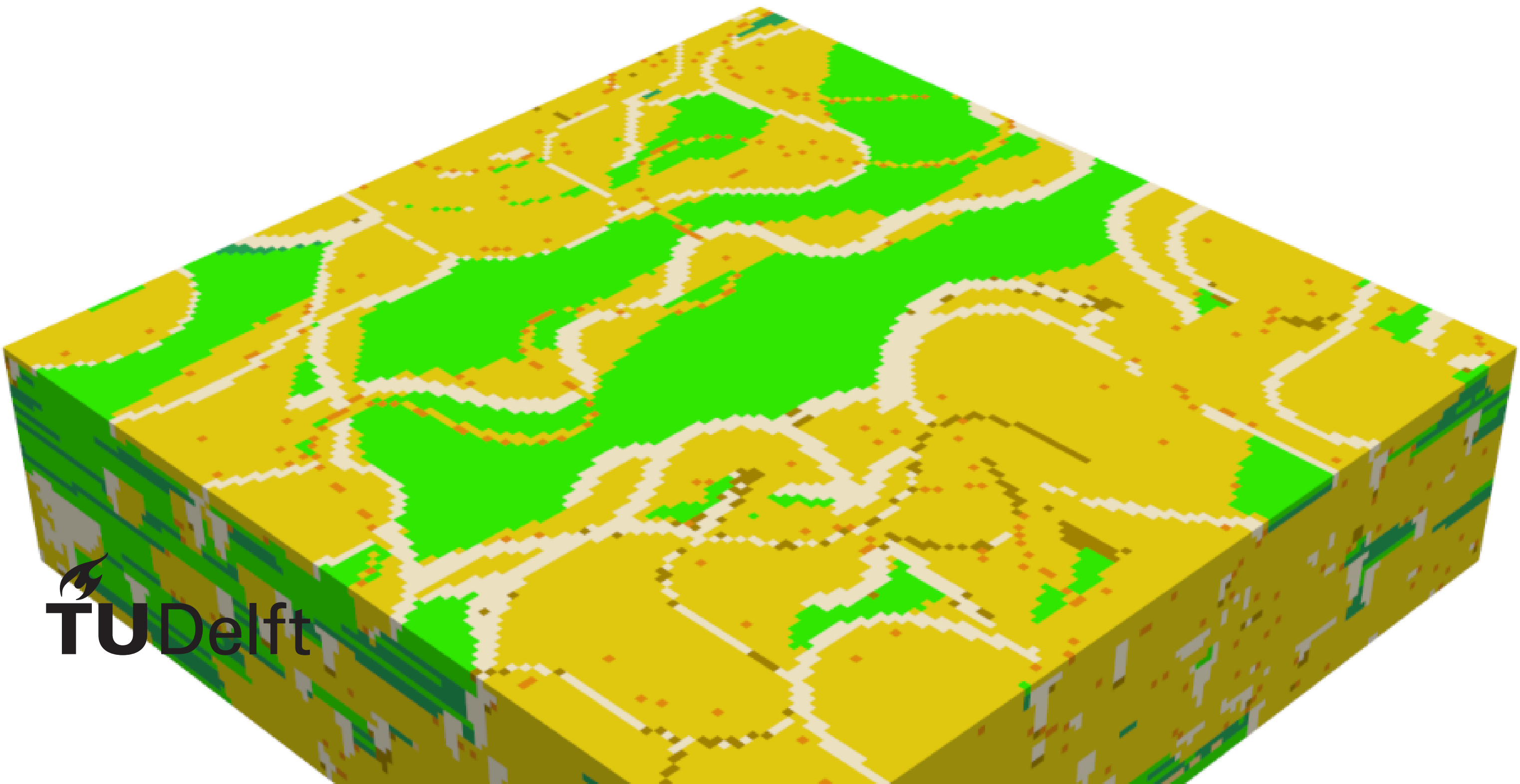
# Evaluating the applicability of Generative Adversarial networks to generate geological facies models

## A Case study for the Delft Area

M.D. Torres Ruhe



# Evaluating the applicability of Generative Adversarial networks to generate geological facies models

A Case study for the Delft Area

by

M.D. Torres Ruhe

to obtain the degree of Master of Science
at the Delft University of Technology,
to be defended publicly on Tuesday January 1, 2013 at 10:00 AM.

| | | |
|---|---|---|
| Student number: | 1234567 | |
| Project duration: | March 1, 2012 – January 1, 2013 | |
| Thesis committee: | Prof. dr. ir. J. Doe, | TU Delft, supervisor |
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*This thesis is confidential and cannot be made public until December 31, 2013.*



TU Delft

# Abstract

Here is the text of my abstract...

# Acknowledgements

I would like to thank...

# List of Abreviations

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**cGAN** Conditional Generative Adversarial Network. 11

**DSST** Delft Sandstone. 3, 4

**FluvGan** Fluvial Generative Adversarial Network. 11

**GAN** Generative Adversarial Network. 11

**TUD** Technische Universiteit Delft. 3

**TvD** True Vertical Depth. 3

**WNB** West Netherlands Basin. 3

# Contents

# 1 Introduction

Sedimentary deposits represent a vital commodity in modern society due to the various resources that can be extracted from them. These resources include minerals, groundwater, geothermal heat, sands and hydrocarbons (**morris2007genesis**; **donselaar2017relation**; **donselaar2015reservoir**; **brekke2001sedi** Moreover, recent studies have investigated storage possibilities for carbon dioxide and hydrogen in this subsurface environment (**zhou2020dynamic**; **heinemann2021enabling**). Beyond the increase in subsurface applications, the world also deals with an increasing population and energy demand. Consequently, the demand for resources hosted within fluvial sedimentary deposits is expected to increase while the distribution of the commodities themselves becomes sparser and of lower quality (**spittler2020modelling**; **calvo2016decreasing**). This highligts the need for a deeper understanding of subsurface environments to enable more precise resource management which can be achieved through various methods such as: data gathering surveys, geological evolution and constructing high-quality 3D geomodels.

3D geomodels are traditionally constructed by engineers and geoscientists to predict reservoir properties and resource distribution (**song2022review**; **royer20153d**). The voxels in these models may be populated by continuous or categorical properties: grain size, porosity & permeability values or facies associations. These predictions are fundamentally based on a foundational geological conceptual model. The conceptual model acts as a hypothesis regarding the most likely rock mass configuration, built entirely from sparse field data (**koltermann1996heterogeneity**). This data sparsity is an unavoidable reality of subsurface exploration, resulting from the exceptionally high costs and extended runtimes required for physical data acquisition. Consequently, engineers develop a working understanding of the subsurface based on limited information. This reliance on limited information drives the continuous need to improve geomodelling techniques. Such improvements aim to computationally bridge the gap between isolated data points and realistic, continuous geological representations.

## 1.1. Problem definition

Despite ongoing advancements in data collection, data scarcity dictates that functional 3D geomodels must rely heavily on subjective interpretations of the underlying geological conceptual model. Because transitioning from these interpretive hypotheses into a digital framework requires applying specific mathematical and spatial algorithms, this workflow inevitably introduces uncertainty at multiple stages, stemming from both the sparse data inputs and the modelling steps themselves (**hoyer2024evaluating**). As a result, precisely measuring its extent is crucial for effective decision-making and can be accomplished with various techniques (e.g. Monte Carlo Simulations), necessitating the creation of a large collection of equally likely 3D geomodels. Creating this extensive ensemble, however, reveals a core drawback in conventional modeling techniques: the significant compromise between computational efficiency and geological precision.To sustain the generation rate needed for extensive uncertainty quantification, traditional methods frequently depend on greatly simplified, weak geological assumptions.

Due to the fact that these approaches are often limited to simple statistical distributions or strict object-oriented rules (**rongier2025atowards**), they find it challenging to replicate the intricate, non-linear architectural relationships present in natural sedimentary formations (**sun2023geological**). As a result, existing workflows compel engineers to decide between models that can quickly quantify uncertainty but do not possess geological realism, or highly precise multi-conceptual models that require too much effort to produce in large quantities (**ENEMARK2019310**).This bottleneck highlights a significant deficiency in contemporary subsurface characterization. A significant need exists for a modeling approach that can thoroughly incorporate complex geological prior knowledge while also ensuring the fast generation rates necessary for ensemble-based uncertainty quantification. As a result, the inquiry comes up regarding how the existing modelling process can be modified to ensure that:

- 1) Previous geological insights and structural intricacies are better employed.

- 2) Uncertainty in the produced models is quantified more precisely and efficiently.

To address the need for both geological realism and rapid ensemble generation, recent research has proposed advanced machine learning algorithms as a highly effective solution. These algorithms, specifically the modern subset known as Deep Learning, possess the unique ability to learn complex, non-linear spatial relationships directly from data. By learning these intricate relationships, Deep Learning can accurately capture the geological complexity and non-stationarity that traditional methods often miss (**rongier2025atowards**). Furthermore, once such a model is fully trained and validated, the generative subset of Deep Learning models can generate a vast ensemble of 3D realizations at a fraction of the time compared to slower process-based models. This rapid generation directly facilitates the large-scale uncertainty quantification demanded by modern subsurface engineering.

While the broader application of machine learning in geomodelling has historical roots (**demyanov2008geomodellin** recent surges in computing power have yielded highly capable 3D Deep Generative Modelling (DGM) architectures, including Variational Auto Encoders (VAEs) and Latent Diffusion Models (**laloy2017inversion**; **bhavsar2024stable**; **song2023gansim**; **federico2025three**). Among these varied architectures, this study specifically focuses on Generative Adversarial Networks Generative Adversarial Networks. While recent literature highlights Latent Diffusion Models as a vanguard for precise hard data conditioning, they inherently suffer from slow inference speeds due to their iterative denoising process. In contrast, Generative Adversarial Networks offer near-instantaneous sampling speeds perfectly suited for large-scale Monte Carlo uncertainty loops, despite well-known training vulnerabilities such as mode collapse. Generative Adversarial Network are therefore selected due to their computational efficiency, established research history in spatial generation, and the wide availability of architectures strictly tailored for geomodelling. However, successfully training such Generative Adversarial Networks (GANs) requires an extensive volume of high-quality data to act as the geological prior. To overcome the inherent scarcity of real subsurface data, process-based forward modelling software, such as FLUMY (**martin-hal-01313232** can be utilized to generate robust synthetic training datasets.

## 1.2. Research gap and opportunity

Utilizing these synthetic datasets allows Generative Adversarial Networks to effectively learn geological priors; however, a critical research gap remains regarding their ultimate deployment. While DGM excels in theoretical environments, recent literature emphasizes an urgent need to transition from idealized, synthetic test cases to practical field applications. As **rongier2025atowards<empty citation>** stated, “We now need to move towards more detailed studies on larger case studies closer to field applications to further assess how the whole workflow performs”. Similarly, **federico2025three<empty citation>** noted that the application (and extension as required) of the overall methodology to real cases should be performed. Addressing this explicit mandate forms the core opportunity of this research. Consequently, the primary goal of this project is to bridge the divide between synthetic training environments and practical field operations by assessing the performance of deep generative modelling in a more complex, real-world application.

**Add: section with the full opportunity for this study**

Testing these models in a real-world application requires targeting an active subsurface system, a role fulfilled in this study by utilizing data from the Delft Sandstone reservoir. This reservoir target must be characterized using a simple geological conceptual model combined with sparse well data. Such sparse well data will serve as the hard conditioning constraints to evaluate how the generative model behaves under tight physical limitations. These limitations, and the resulting fluctuations in uncertainty, will be analyzed by testing the workflow across multiple distinct geological priors. Investigating the interplay between varied priors and sparse conditioning constraints ultimately establishes the framework for the specific objectives of this study.

## 1.3. Objective and research questions

With the research gaps in mind, the main goal of this thesis will be to assess the applicability of using a Generative Adversarial Network to generate realizations of a real-world reservoir by conditioning it to well data. Based on the outcome of this thesis, new insight can be given into the workflow needed to approach applying and evaluating Generative Adversarial Networks for generating subsurface reservoir realizations. Hence, the main research question of this work is:*To what extent can Generative Adversarial Networks be used to create geologically plausible realizations of the Delft Sandstone reservoir while conditioned to sparse real-world data?* To help answer this question, several sub-research questions have been formulated:

1. How should the Delft Sandstone's geology be parameterized in FLUMY to produce a training dataset sufficiently diverse for DGM modeling?
2. Which combination of hyperparameters yields a stable training process?
3. To what degree is the model able to reproduce the complexity inherent in the training data?
4. To what degree does the model generate realizations that are succesfully conditioned to well data?
5. How different do models trained on different geological priors react when sparse data is introduced?

## 1.4. Thesis outline

The remainder of this thesis is structured to systematically address these research objectives. Chapter 2 establishes the theoretical foundation by reviewing the principles of geological modelling, the mechanics of deep generative models, and the specific geological setting of the Delft Sandstone Member. Building upon this theoretical background, Chapter 3 outlines the core workflow, describing the generation of synthetic training datasets using FLUMY and detailing the specific Generative Adversarial Network (GAN) architecture deployed for the reservoir modelling tasks. The empirical outcomes of this workflow are presented in Chapter 4, which evaluates the training process and analyzes the structural and spatial realism of the generated realizations. These results are critically examined in Chapter 5, which contextualizes the findings, addresses the inherent limitations of the proposed approach, and highlights avenues for future research. Finally, Chapter 6 summarizes the main contributions of this thesis and provides definitive answers to the central research questions.

# 2 Background

This chapter establishes the theoretical and geological foundations and reasoning pon which this research is built. It provides a short overview of the geological setting of the Delft reservoir and a literature review of modern geomodelling and machine learning (ML) techniques.Section 2.1 introduces the geological context of the Delft reservoir, detailing its depositional environment and current conceptual model. Section 2.2 explores the state-of-the-art in geological modelling, comparing traditional techniques and their respective limitations. Finally, Section 2.3 provides a general overview of machine learning and deep learning (DL) principles, focusing on their historical evolution and current application to subsurface characterization. This balanced overview is intended to provide both geological and computer science readers with the necessary context to understand the research methodology and results presented in subsequent chapters.

## 2.1. The Delft Geothermal Reservoir

The data utilized in this study is sourced from the Delft geothermal reservoir, situated beneath the Technische Universiteit Delft (TUD) Campus in the Netherlands. As of February 2026, a geothermal doublet is actively producing heat from this reservoir (**beerda2026geothermie**). A doublet system consists of a two-well setup: a producer well extracts warm formation water, and an injector well returns the cooled water to the reservoir. At the Delft site, the producer well (**DEL-GT-01**) was drilled to a True Vertical Depth (TvD) of approximately 2,337 m, while the injector well (**DEL-GT-02-S2**) reached a TvD of approximately 2,100 m (**vardon2024endofwell**). The project serves both commercial and research objectives, acquiring extensive geophysical logs and core data, supplemented by surface and downhole monitoring instrumentation (**vardon2020geothermal**; **vardon2024endofwell**). Despite the high data density, the cored section is restricted to the first 80 m of the **DEL-GT-01** well (approximately one-fourth of the reservoir thickness), while **DEL-GT-02-S2** contains no cored sections. The primary target is the Delft Sandstone (DSST), a member of the Lower Cretaceous Nieuwerkerk Formation in the West Netherlands Basin (WNB) (**donselaar2015reservoir**).

### 2.1.1. Geology of the Delft Reservoir

The WNB is a 60 km wide transtensional basin in the southwest of the Netherlands, characterized by NW-SE trending structural highs and depressions (**ziegler1990geological**; **donselaar2015reservoir**). The WNB entered an active rifting stage during the Middle Jurassic (**bodenhausen1981habitat**; **den1996fluviomarine**; **willems2020geology**), leading to the formation of several half-grabens filled with terrestrial sediments sourced from neighboring regions (**den1996fluviomarine**). The progressive filling of these half-grabens coincided with a change in the depositional environment as a transgressive shoreline moved the system from fluvial to marine. The Nieuwerkerk Formation comprises three members: the Alblasserdam, the DSST, and the Rodenrijs Claystone.

The Alblasserdam member was deposited in a fluvial environment characterized by stacked channels and floodplain deposits (**devault2002tectonostratigraphy**). The DSST directly overlays the Alblasserdam and consists of thick, stacked fluvial-deltaic channel sandstones with high net-to-gross values, oriented NW-SE (**willems2020geology**; **mijnlieff2020introduction**; **bruining2023sedimentological**; **groenenberg2010targeting**). The DSST is in turn overlain by the Rodenrijs Claystone, an organic-rich lagoonal deposit (**willems2020geology**; **devault2002tectonostratigraphy**; **bruining2023sedimentologi**

Based on the sedimentological analysis of the **DEL-GT-01** core and **DEL-GT-02-S2** sidewall samples, three primary facies associations (FAs) have been identified (**bruining2023sedimentological**):

- **FA1: Fluvial and distributary channel deposits.** Primarily light grey, fine- to medium-grained sandstones featuring low-angle cross-bedding and ripple lamination, with unit thicknesses ranging from 5 to 80 m.

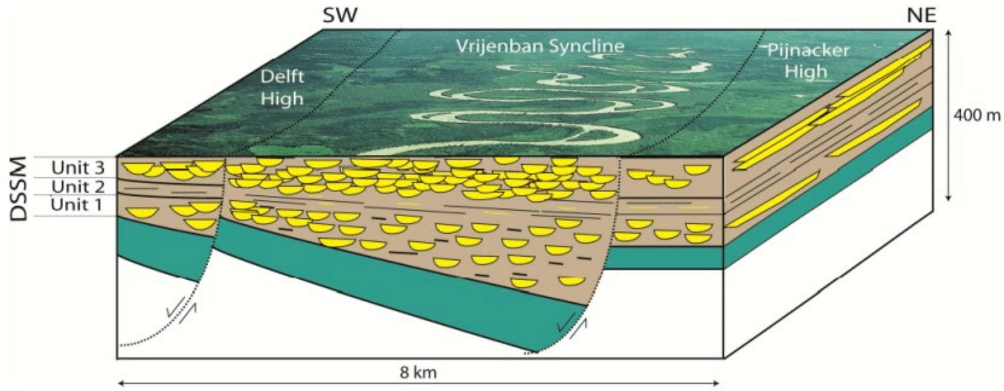



Figure 2.1: Depositional models for the DSST Member with three identified units. Accomodation space is increasing towards the NE (**donselaar2015reservoir**)

- **FA2: Crevasse splays and levee deposits.** Composed of moderately sorted, very fine- to medium-grained sandstones interbedded with dark grey siltstones, with thicknesses ranging from 1 to 5 m.
- **FA3: Overbank deposits and paleosols.** Consists of dark grey siltstones, claystones, and very fine-grained sand deposited in low-energy environments. This association is characterized by abundant organic material, slickensides, and siderite nodules.

The specific depositional environment of the DSST remains a subject of scientific discussion. It has been attributed to both fluvial-meandering (**donselaar2015reservoir**; **vondrak2018reservoir**; **willems2020geology**) and fluvio-deltaic systems (**van1993stratigraphic**; **weert2024multiple**; **bruining2023sedimentological**). Recent analysis by **bruining2023sedimentological<empty citation>** suggests that the cored section represents a deltaic environment. The observed facies associations can be attributed to a upper to lower plain delta with multi-stacked fluvial/distributary channels, single channels, crevasse splays, levees and ovrbank fines.

### 2.1.2. Deltaic Depositional Environment

Deltaic environments form at the land-sea interface where river systems supply clastic sediments. Delta morphology is governed by the relative influence of fluvial, wave, and tidal processes (**kolb1966depositional**). The subaerial delta plain is typically divided into two domains (**kolb1966depositional**; **heldreich2017challenges**):

- **Upper Delta Plain:** Lies above the limit of saltwater intrusion. Sedimentation is driven by distributary channel migration, overbank flooding, and crevassing. Environments include meandering and braided channels, lacustrine fills, and floodplains.
- **Lower Delta Plain:** Subject to river-marine interaction and tidal influence. Channels are more numerous and exhibit bifurcating or anastomosing patterns. Interdistributary bays, bay fills (crevasse splays), and marshes dominate the landscape.

In the DSST, FA1 features organic material and siltstones indicating slackwater periods, while the lack of bioturbation suggests high sedimentation rates. The fine-grained nature of FA2 and FA3 reflects more distal positions on the delta plain. Siderite nodules in FA3 indicate anoxic, waterlogged conditions (**samuel1959some**), while soft-sediment deformation suggests rapid loading of mudstones over saturated sands. These features collectively indicate that FA3 represents deposition within a delta plain, likely in an interdistributary bay or a pro-delta environment.

## 2.2. Geological Modelling

The primary objective of 3D reservoir modelling is to provide a representative volume of the subsurface for flow simulation and decision support. These models are dynamic, requiring updates as new data becomes available to refine predictions over the reservoir's lifetime (**emerick2025ensemble**; **ringrose2016reservoir**). Existing techniques are generally categorized into three types (**koltermann1996heteroge**

- **Pixel-based Models:** Use spatial statistics (e.g., variograms) for cell-by-cell property assign-

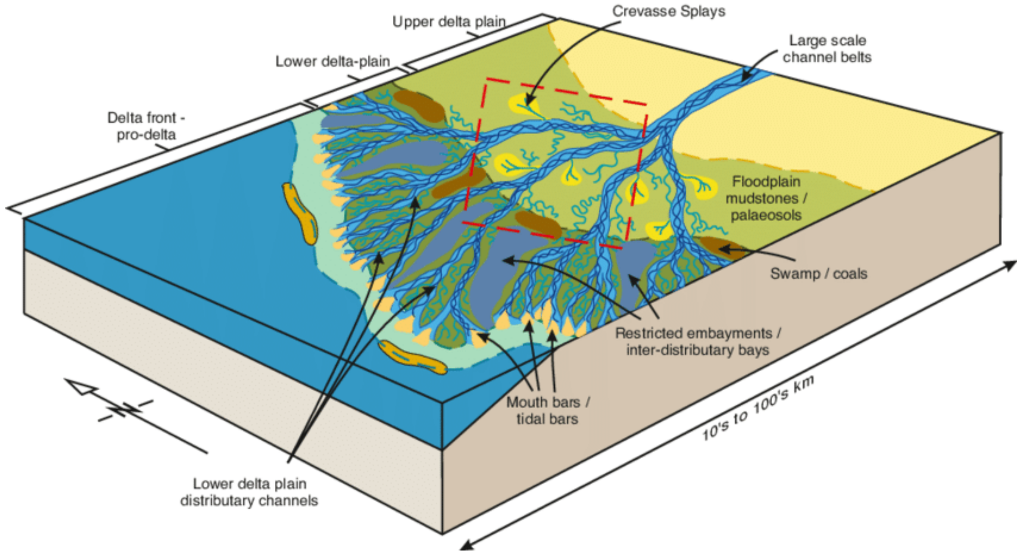



Figure 2.2: Depositional model of the Mungaroo Formation, illustrating the geometries and domains of a fluvial-deltaic system (**heldreich2017challenges**).

ment. They are efficient and honor dense well data but often fail to capture complex connectivity, resulting in pixelated realizations.

- **Object-based Models:** Directly place geometric features (e.g., channels) based on geological rules. They offer high visual realism but struggle to honor dense or closely spaced well data.
- **Process-based Models:** Simulate the processes behind sediment transport and deposition. They provide the highest degree of realism but are computationally expensive and difficult to condition to specific hard data like well logs.

| Attribute | Geostatistical | Object-Based | Process-Based |
|---|---|---|---|
| **Geological Prior** | Variograms, training images. | Geometric shapes and rules. | Physical boundary conditions. |
| **Realism** | Low to Moderate. | Moderate to High. | Very High. |
| **Key Advantages** | Computational speed, well conditioning. | Realistic connectivity. | Physical consistency. |
| **Key Disadvantages** | Lack of geological realistic shapes. | Difficult conditioning. | High computational cost. |

Table 2.1: Comparison of spatial modelling techniques in geomodelling.

While traditional pixel-based geomodelling relies heavily on two-point spatial statistics, such as variograms, to distribute properties, these methods often fail to capture the complex, multipoint connectivity inherent in natural geological systems. Figure Machine learning introduces a paradigm shift by moving away from user-defined spatial rules. Instead of requiring an engineer to pre-define the shape of a channel or the range of a variogram, ML algorithms learn these intricate, non-linear spatial relationships directly from data. This makes ML exceptionally well-suited for geomodelling, where the relationships between sparse hard data (e.g., well logs) and the broader geological architecture are too complex to be captured by standard statistical interpolation.

## 2.3. Machine Learning

Machine learning (ML) involves the development of algorithms that learn patterns from data to perform specific tasks without explicit programming (**goodfellow-et-al-2016**). This foundational concept of

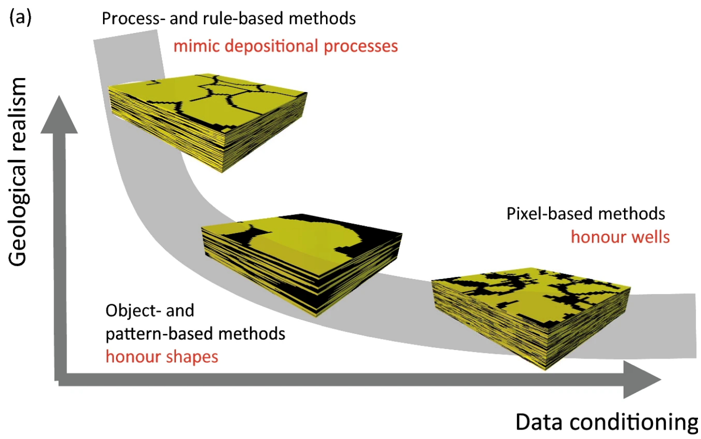



Figure 2.3: Simple overview of different geological modelling techniques compared to their ability to generate geological plausible data and honour well data (**manzocchi2026general**). Where process- and rule- based methods are great at generating geological plausible samples but strugle with honouring to well data, pixel-based methods are opposite in nature. Object based models offer a best of both worlds solution.

learning without explicit programming has evolved significantly since **samuel1959some<empty citation>** introduced the term, expanding into a wide array of computational techniques. These diverse techniques are fundamentally categorized based on their training approach into two primary domains:

- **Supervised Learning:** The algorithm is trained on a labeled dataset containing explicit input-output pairs.
- **Unsupervised Learning:** The algorithm identifies hidden patterns within unlabeled data without prior knowledge of the desired output.

Within these two broad categories, algorithms are deployed to solve common tasks such as classification, regression, clustering, and generation. Applying these generative tasks to the subsurface highlights a severe limitation in traditional pixel-based geomodelling, which relies heavily on simple two-point spatial statistics (e.g., variograms) to distribute properties. Because these simple statistics often fail to capture the complex, multipoint connectivity inherent in natural geological systems, ML introduces a necessary paradigm shift. This shift moves the modelling workflow away from strict, user-defined spatial rules. Instead of requiring an engineer to pre-define the range of a variogram, ML algorithms learn intricate, non-linear spatial relationships directly from the data. Learning directly from data makes ML exceptionally well-suited for geomodelling, where the relationships between sparse hard data (e.g., well logs) and the broader geological architecture are simply too complex to be captured by standard statistical interpolation. The capacity to uncover such complex, non-linear relationships in vast datasets is largely driven by modern Artificial Neural Networks (ANNs) (**lecun2015deep**). Inspired by the biological structure of the brain, these ANNs consist of interconnected artificial neurons that process information across multiple computational layers (**goodfellow-et-al-2016**)

While ANNs and traditional machine learning algorithms excel at analyzing tabular petrophysical data for various subsurface tasks, they inherently lack actual geological reasoning skills. Without such reasoning to naturally make sense of areas with low data quality, these algorithms are restricted to inferring abstract features directly from their training data (**schaaf2019quantification**). Inferring these features in standard ML workflows necessitates manual feature extraction, a process where engineers must explicitly define the relevant geological parameters prior to training. This strict reliance on manual feature extraction creates a critical limitation when attempting to scale these algorithms to 3D geomodels. Within these high-resolution 3D geomodels, which typically contain millions of voxels, manually defining multi-scale spatial features ultimately becomes computationally impractical.

Deep Learning (DL) overcomes this limitation through automatic feature extraction. By utilizing multi-layered networks, DL architectures natively process high-dimensional spatial grids, bypassing the manual extraction bottleneck. For subsurface characterization, this means a DL model can ingest raw 3D conceptual models or process-based simulations and autonomously identify the underlying geological rules. This capability to scale with high-dimensional data makes DL the premier mathematical framework for modern reservoir modelling.

### 2.3.1. Deep Learning and Inductive Biases

Deep learning (DL) is a subset of ML focusing on multi-layered neural networks that represent data with multiple levels of abstraction (**goodfellow-et-al-2016**; **lecun2015deep**). Achieving these levels of abstraction in high-dimensional spaces introduces a fundamental challenge known as the "curse of dimensionality." This curse of dimensionality describes the exponential increase in data required as the spatial dimensions grow, making high-dimensional estimation computationally challenging. To overcome this computational challenge and help models generalize from limited data, engineers implement specific architectural choices known as inductive biases (**prince2023understanding**). A premier example of such an inductive bias is the foundational architecture used for processing spatially structured data: the Convolutional Neural Network (CNN).

CNNs leverage spatial inductive bias to become highly effective for 2D and 3D geomodelling tasks. Executing these spatial tasks relies on a core architectural innovation known as the convolutional layer. This convolutional layer replaces the fully connected neurons of a standard ANN with a localized set of learnable filters, commonly referred to as kernels. These multi-dimensional kernels operate by systematically sliding across the input data in discrete increments dictated by a predefined step size, or stride. This sliding stride is frequently combined with padding, a technique that adds artificial zero-value borders to the input to ensure the entire spatial region is comprehensively analyzed. Analyzing this spatial region involves computing a local dot product between the kernel weights and the input values, a mathematical operation visually illustrated in Figure 2.4.

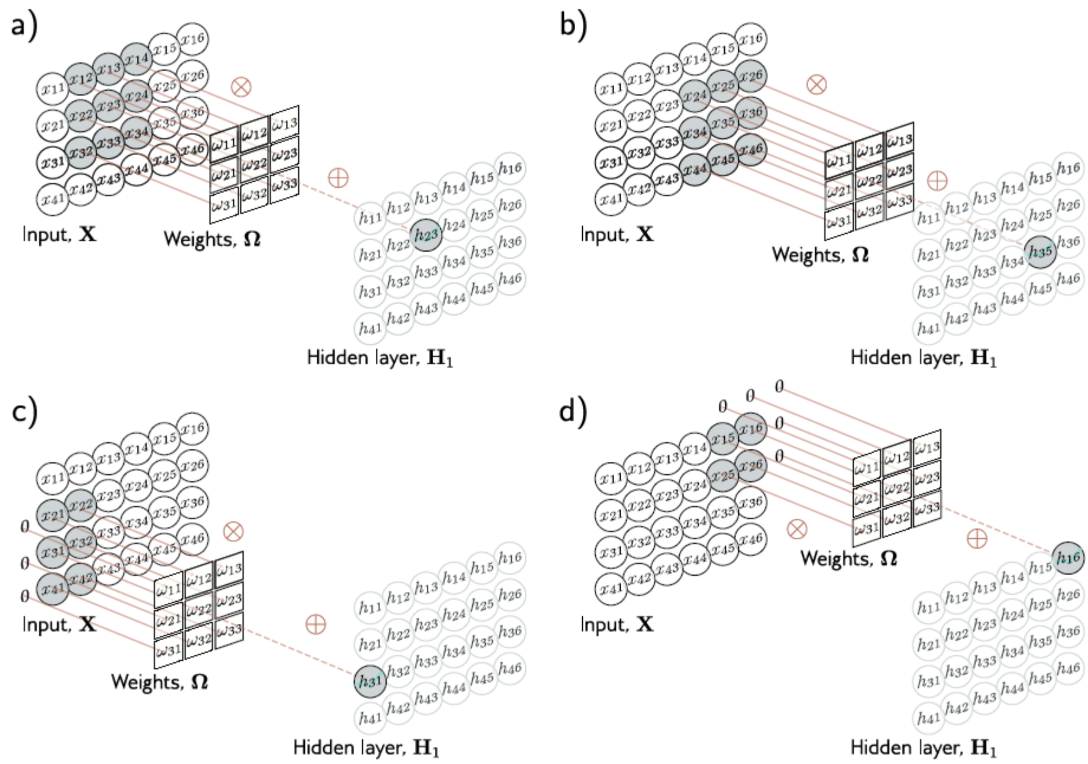



Figure 2.4: Example of a 2D convolutional layer with kernel size of 3x3 and a padding of 1. Each output $h_{ij}$ is composed of a weighted sum of the 3x3 nearest inputs on the previous layer (**prince2023understanding**). a) Here, the output h23 (shaded output) is a weighted sum of the nine positions from x12 to x34 (shaded inputs). b) Different outputs are computed by translating the kernel across the image grid in two dimensions. c–d) With zero-padding, positions beyond the image's edge are considered to be zero.

The mathematical result of this operation is a new, filtered spatial representation known as a feature map. This feature map is typically downsampled by a subsequent pooling layer, which condenses the spatial dimensions while preserving the most prominent information. Preserving this information across a continuous stack of convolutional and pooling layers allows the network to progressively map low-dimensional input data into high-dimensional, abstract features. At the base of this feature mapping process, the initial layers capture simple, low-level spatial patterns, such as discrete edges or sharp facies boundaries. By mathematically combining these simple low-level edges in deeper layers, the network learns to recognize highly complex, high-level geological architectures, such as the overall sinuosity of a fluvial channel. Learning these complex high-level architectures is computationally efficient because the exact same kernel is applied universally across the entire grid, creating an inherent advantage known as translation invariance. This translation invariance ensures that if the network learns to identify a channel structure in one corner of the reservoir, it can accurately recognize that same structure anywhere else in the model, drastically reducing computational overhead and preventing overfitting. Further preventing this overfitting requires another critical component of the DL workflow: the strict use of validation and testing protocols. These validation protocols ensure that the hierarchical patterns learned by the network are robust and safely applicable to unseen data. Applying these robust, learned spatial patterns to unseen subsurface data forms the fundamental basis of modern Deep Learning (DL)-driven geomodelling workflows.

### 2.3.2. Geomodelling with Deep Learning

Modern DL-driven geomodelling workflows have evolved significantly, transitioning from simple pixel-based interpolations to the generation of highly sophisticated spatial representations. Generating these sophisticated representations relies on the ability of DL to directly learn complex geological features from large datasets or process-based simulations. Learning directly from such simulations bridges the historical gap between simplistic descriptive statistics and true geological realism (**laloy2017inversion**). To achieve this realism, CNNs have proven particularly successful in capturing the complex, non-stationary architectures of fluvial channels and facies distributions (**sun2023geological**). Capturing these intricate architectures enables the rapid generation of stochastic realizations that successfully honor both the global geological concept and local, sparse conditioning data. Honoring these dual constraints while generating massive ensembles of stochastic realizations fundamentally enhances subsurface uncertainty characterization (**song2023gansim**). This unique capacity for rapid, constrained generation is primarily driven by specialized neural network frameworks, most notably generative architectures.

These generative architectures that are using CNNs, are best known for their ability to generate realistic images. The most-well known models include VAEs, GANs and Latent Diffusion Models. Though they are all diffent in terms of architecture, some core concepts remain similar. Generative models are designed to learn and approximate the complex joint probability distribution of the training data whereby elements are generated based on previously generated values (**he2025generative**). Moreover, these models use a latent space, designed to either store the encoded trainin data (VAEs) or be used to scale to an image through various upsampling steps (GANs and Latent Diffusion Models) (**prince2023understanding**).

### 2.3.3. Generative Adversarial Networks

Generative architectures, specifically Generative Adversarial Networks (GANs), achieve complex spatial generation through a competitive framework between two distinct neural networks (**goodfellow2014generative**). These two networks are defined as a Generator ($\mathbf{G}$) and a Discriminator ($\mathbf{D}$). The generator is tasked with creating synthetic spatial samples from random noise, while the discriminator is simultaneously tasked with distinguishing these synthetic samples from actual real-world training data. The continuous competition to outsmart one another drives a minimax game, which is mathematically expressed as:

$$\min_{\mathbf{G}} \max_{\mathbf{D}} V(\mathbf{D}, \mathbf{G}) = \mathbb{E}\mathbf{x} \sim p\text{data}(\mathbf{x})[\log \mathbf{D}(\mathbf{x})] + \mathbb{E}\mathbf{z} \sim p\mathbf{z}(\mathbf{z})[\log(1 - \mathbf{D}(\mathbf{G}(\mathbf{z})))] \tag{2.1}$$

In this equation, $\mathbf{x}$ represents the real data, and $\mathbf{z}$ represents a random latent vector sampled from a prior distribution $p_{\mathbf{z}}$. The first term represents the discriminator successfully recognizing real geological data, while the second term represents the discriminator successfully catching fake data generated by $\mathbf{G}(\mathbf{z})$. During training, the discriminator seeks to maximize this overall value function to perfect its detection abilities, while the generator seeks to minimize it to successfully fool the discriminator.

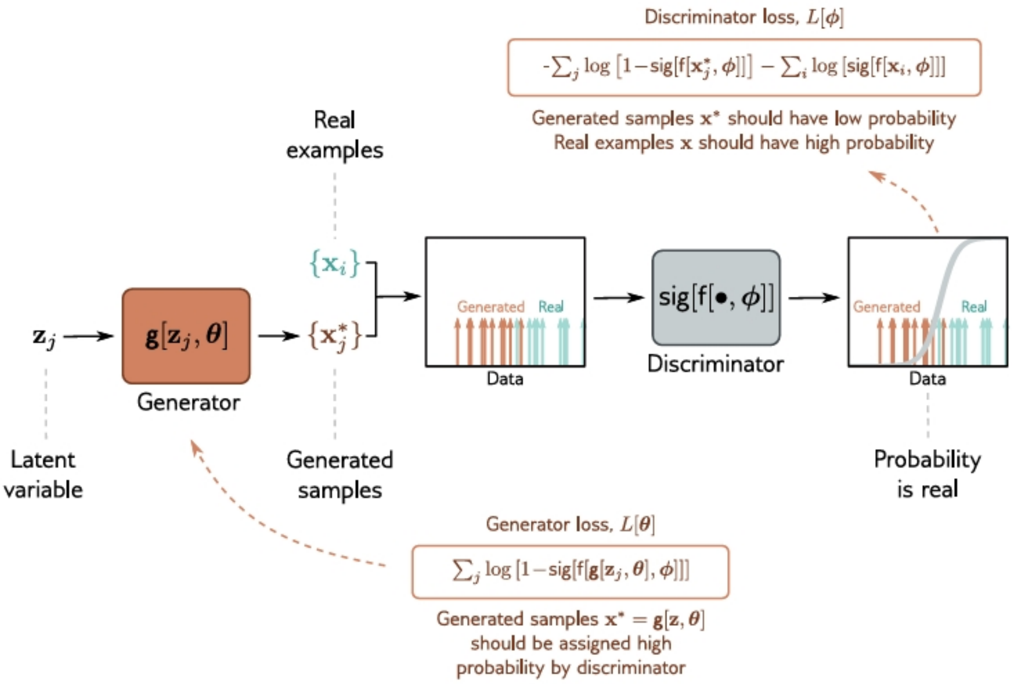



Figure 2.5: Basic overview of a standrd GAN architecture with a Generator, constructing fake samples from a random latent variable $z_j$ and a Discriminator deciding which of the two samples is the real sample (**prince2023understanding**).

An overview of these steps, describing a single iteration of the model, is shown in figure 2.5. By successfully fooling the discriminator, the generator ultimately learns to produce high-fidelity geological realizations without requiring any explicit, hard-coded spatial rules.

### GAN Inversion

Figure 2.5 illustrates the generation of high-fidelity, unconstrained realizations, which is a powerful capability; however, practical reservoir modelling requires these models to honor sparse physical constraints like hard well data. Hard well data constraints are typically honored through optimization techniques (**rongier2025btowards**; **bhavsar2024stable**; **chan2019parametric**). Several techniques have been proposed including applying inference networks (**bhavsar2024stable**; **chan2019parametric**), Gaussian Mixture modelling (**bhavsar2024stable**), laten space optimization and inversion (**rongier2025btowards**) Where inference and GMM (ADD EXPLANANTION ON THESE TECHNIQUES) GAN inversion operates strictly within the model's latent space after the training phase is complete. Completing the training phase allows the algorithm to iteratively search and optimize this latent space to identify a specific latent vector z. This specific latent vector z generates a 3D volume perfectly matching the sparse well data. Matching the sparse well data securely conditions the generative model to real-world subsurface observations. Moreover, a second proposed technique by **rongier2025btowards<empty citation>** is pivotal tuning, a technique originally designed to fine-tune the generator to comply a set latent space (**roich2022pivotal**).

# 3 Methodology

This chapter describes the project workflow, beginning with the generation of 3D synthetic datasets used for training, validating, and testing the models (Section 3.1). Section 3.2 details the GAN framework, including the network architectures, the facies representation strategy, and the loss functions employed during training. Finally, Section 3.3 introduces the metrics and simulation methods used to evaluate the geological realism and practical utility of the generated realizations.

The primary real-world constraints for this project are derived from the physical well logs of the DEL-GT-01 and DEL-GT-02-S2 geothermal wells. **bruining2023sedimentological<empty citation>** interpreted the cored sections of these wells, translating the raw physical logs into distinct, combined facies associations. The facies associations along the wellbore presented in figure B.1. The specific categorical nature of this well data serves as the link between the synthetic generative models and practical field application, driving two key methodological decisions in this project.

1. Because the hard data is classified by structural facies associations rather than exact, continuous petrophysical properties, it dictated the decision to train the generative models on discrete facies classes instead of continuous grain size distributions.
2. The resolution of these well-log interpretations, which synthesize the complex subsurface into three primary facies associations, directly necessitated the subsequent grouping of the highly detailed synthetic environments into an upscaled 3-facies dataset.

By aligning the model's categorical resolution with the exact structure of the available well data, it ensures that the generative framework can be practically conditioned on and evaluated against actual physical subsurface evidence.

## 3.1. Datasets

In order to test the applicability of GANs, new datasets had to be generated since training deep learning algorithms for 3D geomodelling requires a substantial volume of high-fidelity training data. The new datasets were generated using FLUMY (**martin-hal-01313232**), a process-based stochastic modelling software designed to reproduce meandering channelized reservoirs. Based on sedimentological rules, the software reproduces facies type, grain size and deposition time for each voxel in th 3D volume.

### 3.1.1. Dataset Descriptions

To test the GAN in different scenario', three distinct geological conceptual models were curated to capture a range of environments for the Delft Sandstone reservoir. The setting specific input parameters are shown in Table 3.1. The hypothesized depositional environments are split across three discrete

| Parameter | Setting 1 | Setting 2 | Setting 3 |
|---|---|---|---|
| Net-to-Gross (%) | 67 | 67 | 67 |
| Max. Channel Depth (m) | 5 | 6 | 7 |
| Max. Sand Body Extension | 80 | 100 | 120 |

Table 3.1: NEXUS input parameters for the three training datasets.

suites, ranging from a Lower plain to Upper plain deltaic characteristics in terms of increasing channel depth and wider sand bodies. Beyond these constraints, all generated models share a uniform reservoir scale with a total grid resolution fixed at 32x128×128 voxels. In order to allow for well placement in the samples, the final dimension should encompass an area of at least 2000x2000m as the **DEL-GT-01** and **DEL-GT-02-S2** have an 1500m spacing between them. Establishing this specific grid resolution combined with a standard voxel dimension of 20 m×20 m×1 m means that the total simulated domain covers a volume of 2560 m×2560 m×32 m. Every single voxel within this acute 32 m vertical interval is ultimately encoded with 1 of the 14 distinct facies (see B.1) available in FLUMY, of which only 9 facies actually occured in the sample ensemble due to the settings specific parameters.

### 3.1.2. Sample preprocessing

To comprehensively assess the capabilities of GANs, the synthetic training datasets are prepared at two distinct categorical resolutions. These two distinct resolutions consist of a high-fidelity 9-facies dataset and an upscaled 3-facies dataset. The high-fidelity 9-facies dataset preserves the detailed depositional environments generated by FLUMY, serving as a baseline to evaluate the model's pure generative capacity and its ability to replicate highly complex spatial distributions. While evaluating this generative capacity is crucial, practical field application requires conditioning the model to actual subsurface observations. These actual subsurface observations, derived from the cored section of the **DEL-GT-01** well, are strictly classified into three primary facies associations (Section 2.1.1). Because the hard well data only resolves to three distinct associations, the original FLUMY depositional environments must be grouped accordingly to create the upscaled 3-facies dataset. This grouping process aggregates the detailed environments into broader categories, as detailed in Table 3.2.

| Value | Facies Key | Color | Description |
|---|---|---|---|
| 1 | Channel & Point Bar (FA1) | | Grouped proximal deposits: Channel Lag (1), Point Bar (2), Sand Plug (3), and Crevasse Splay I (4). |
| 2 | Levee & Crevasse (FA2) | | Grouped intermediate deposits: Crevasse Channel (5), Crevasse Splay II (6), and Levee (7). |
| 3 | Overbank Fines (FA3) | | Grouped distal deposits: Overbank (8), Mud Plug (9), Hemipelagic Plug (10), Wetland/Paleosol (11), Draping (12), and Pelagic (13). |

Table 3.2: Overview of the upscaled 3-facies grouping used for the generative modelling, aggregated from the original 13 FLUMY depositional environments (See Table **??** for the original facies).

Beyond this grouping process, the categorical nature of both the 3-facies and 9-facies datasets necessitates a final preprocessing step known as one-hot encoding (**goodfellow-et-al-2016**). One-hot encoding is critical because standard neural networks process continuous numerical gradients and might otherwise incorrectly interpret integer facies labels as having a mathematical relationship. This misinterpretation is prevented by transforming the original 3D volume into an expanded 4D tensor, where each class is assigned its own dedicated spatial channel. Each spatial channel indicates the presence or absence of a specific facies using values of 1 and -1 (See appendix in C.1).

## 3.2. Fluvgan Framework

### 3.2.1. Model Architecture

The GAN architecture used in this project is based on the Fluvial Generative Adversarial Network (FluvGan) model, introduced by **rongier2025afluvgan<empty citation>**. Thie architecture adapts the BigGAN framework (**brock2018large**) to manage the complex connectivity and high resolution necessary for realistic reservoir models. Where **rongier2025atowards<empty citation>** used a 2-channel approach for their model, one for grain size and one for deposition time, the model used in this project comprises of either 3 or 9 channels, where each channel represents another facies based on the one-hot encoding strategy. The deposition time channel was not used in this project.

The architecture of the model is comprised of a Generator (**G**) and a Discriminator (**D**) built with 3D Convolutional blocks. These blocks enable the generator to transform a, in this case, 100-dimensional latent vector $\mathbf{z}^*$ into a $C \times 32 \times 128 \times 128$ volume. The volume begins as a $1024 \times 4 \times 4 \times 4$ feature map, created via a 3D transposed convolution, and is then expanded by five residual upscaling blocks. The final spatial resolution is processed through a Tanh activation, ensuring the $C$ facies channels are

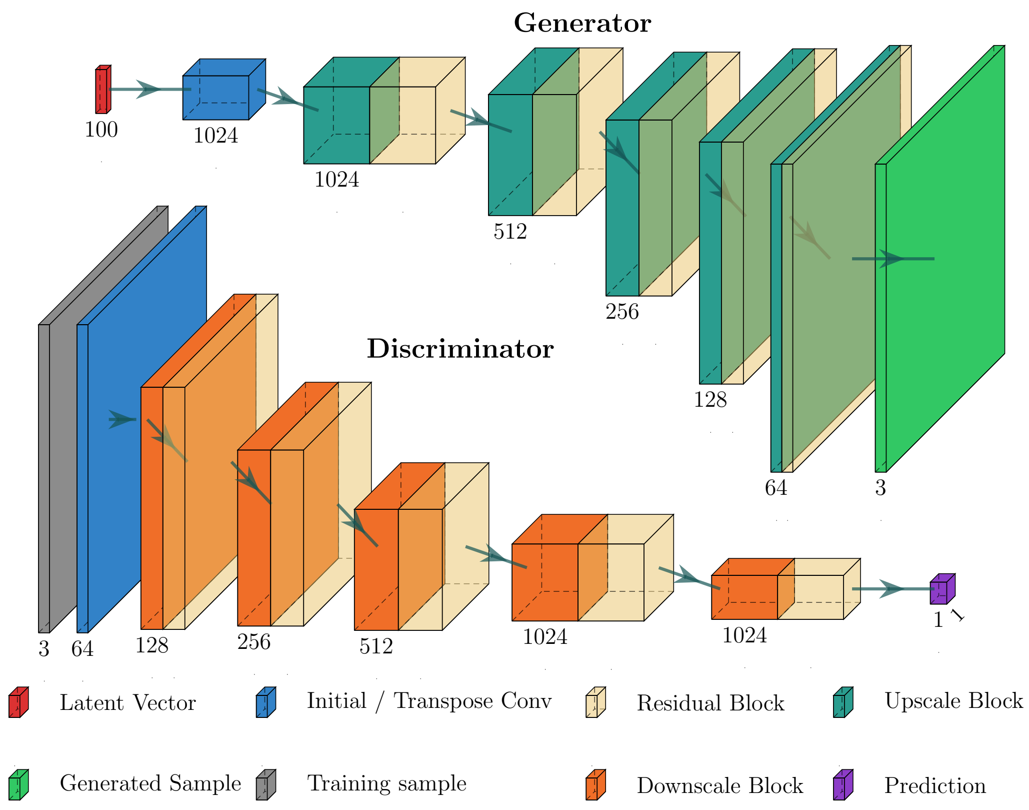



Figure 3.1: **Top)** The Generator initiates from a 100-dimensional latent vector and expands spatial volume through successive upscale and residual blocks. During this process, feature channel density is progressively reduced from 1024 down to 3 channels to construct the final generated volume. **Bottom)** The Discriminator mirrors this pathway by receiving a 3-channel sample (training or generated) and contracting spatial dimensions through downscale blocks. To retain high-level structural patterns during downsampling, feature channels are steadily increased from 3 up to 1024 before compressing into a single purple block representing the final prediction scalar.

scaled scaled from -1 to 1. Finally, an argmax function scales the one hot encoded tensor back to integers, ensuring that a 32x128x128 volume inhabited by class numbers remains. Every generated voxel is then evaluated by the discriminator, acting as a binary classifier to reduce the $C \times 32 \times 128 \times 128$ input to a single scalar logit. This logit is produced through five residual downscaling blocks that iteratively reduce spatial resolution while increasing the channel count to 1024. A simpel visual representation of the model can be seen in figure 3.1. In total, the generator involves roughly 11.6 million trainable parameters while the discriminator has about 2.0 million trainable parameters.

During training, the the Generator and discriminator are updayed with each iteration according to the loss function, which in this case was the BCE Loss

During the adversarial training process, the weights of both the Generator (**G**) and the Discriminator (**D**) are iteratively updated according to a loss function. The fundamental objective of this loss function is to quantify the networks' performance, measuring how accurately the discriminator distinguishes real geological data from synthetic samples, and conversely, how effectively the generator produces data that fools the discriminator. For this architecture, the Binary Cross-Entropy (BCE) loss was chosen as the optimization metric. This choice is directly motivated by the model's one-hot encoding strategy. Because the generator outputs $C$ distinct channels—where each channel represents a specific facies—the spatial generation is effectively transformed into a *multi-channel binary classification task*. For every voxel, the network predicts the independent probability of a specific facies being present or absent. BCE is optimally suited for this structure because it independently evaluates these probabilities, penalizing the pixel-wise deviations between the generated probabilities ($\hat{y}_i$) and the true binary labels

($y_i$). Mathematically, evaluated across $N$ voxels, this is defined as:

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right] \tag{3.1}$$

During adversarial training, instability often manifests as exploding logarithmic losses when either the Generator **G** or Discriminator **D** becomes too confident in their predictions. Furthermore, poorly calibrated networks are highly susceptible to severe mode collapse, where the generator's stochastic capability is restricted. Besides the exact model architecture, other hyperparameters exist that influence how the model behaves during training. To investigate the optimal model architecture and hyperparameters that yield stable training and diverse geological outputs, a systematic hyperparameter tuning study was conducted.

### Hyperparameters

Hyperparameter tuning involves testing various combinations of network settings to evaluate their impact on model performance. The primary objective is to identify an optimal configuration that minimizes the discrepancy between the generated models and the input data, while actively preserving spatial ensemble diversity.To achieve this, baseline parameters (such as the optimizer and batch size) were kept constant (see table 3.3), while a specific set of architectural and training hyperparameters were varied stepwise. The complete search space for this hyperparameter study is outlined in Table 3.4.

| **Hyperparameter** | **Fixed Value** | **Description** |
|---|---|---|
| Optimizer | Adam ($\beta_1 = 0, \beta_2 = 0.999$) | Standard optimization algorithm. |
| Batch Size | 64 | Number of samples processed per iteration. |
| Kernel Size | $3 \times 3 \times 3$ | Dimensions of the 3D Convolutional filters. |

Table 3.3: Fixed baseline configurations for the Fluvgan architecture optimization study.

| **Hyperparameter** | **Search Space / Value** | **Description** |
|---|---|---|
| Learning Rate **G** ($\alpha_{\mathbf{G}}$) | $10^{-3}, 10^{-4}$, Dynamic | Generator step size during gradient descent. |
| Learning Rate **D** ($\alpha_{\mathbf{D}}$) | $3 \times 10^{-3}, 10^{-2}$, Dynamic | Discriminator step size; testing a two-timescale update rule. |
| num iter **D** | 1, 2 | Number of consecutive Discriminator updates per Generator update. |
| Discriminator Penalty | None, 1, 16, 32 | Gradient penalty applied to stabilize Discriminator confidence. |
| Double Conv Blocks | On, Off | Architectural toggle for adding secondary convolutional layers. |
| Double ResBlocks | On, Off | Architectural toggle for adding secondary residual blocks. |
| Training Samples | 10,000, 20,000 | Total volume of synthetic baseline data fed to the network. |

Table 3.4: Hyperparameter search space for the Fluvgan architecture optimization study.

The Learning rates $\alpha_{\mathbf{G}}$ and $\alpha_{\mathbf{D}}$ dictate how magnitude of the network's weights update while the num iter **D** refers to the discriminator update frequency compared to the generator. The Discriminator Penalty serves as a vital regularizer to constrain gradient norms, ensuring the discriminator's confidence does not overpower the generator early in training. Lastly, the architectural toggles (Double Conv and Residual Blocks) paired with data scaling systematically isolate whether network capacity or dataset volume is primary in preserving the necessary spatial and structural diversity of the generated fluvial systems.

### 3.2.2. Optimization & Inversion

While the 9-facies models are utilized to evaluate unconstrained generative performance, the 3-facies models are deployed to test practical conditioning against the **DEL-GT-01** well data. To successfully condition this generative model to real-world observations, the inversion workflow (introduced by (**rongier2025btowards**)) relies on a two-step mechanism: Latent Space Optimization and Pivotal Tuning (**roich2022pivotal**).

First, Latent Space Optimization searches for the optimal random noise vector **z** that best approximates the well data while keeping the pre-trained generator's weights frozen. This iterative search is driven by a composite loss function designed to satisfy two simultaneous objectives: honouring the hard conditioning well data and preserving global geological realism. The total optimization loss ($\mathcal{L}_{total}$) combines a context loss ($\mathcal{L}_{context}$) representing the well mismatch, and a prior loss ($\mathcal{L}_{prior}$) evaluated using the Discriminator's Binary Cross-Entropy (BCE) metric:

$$\mathcal{L}_{total} = \mathcal{L}_{context} + \lambda \mathcal{L}_{prior} \tag{3.2}$$

Here, $\mathcal{L}_{context}$ explicitly penalizes deviations between the generator's output channels and the true one-hot encoded well observations, utilizing distance-based thresholding to localize the penalty. Simultaneously, $\mathcal{L}_{prior}$ ensures the generated sample remains mathematically indistinguishable from real data. By feeding the updated generation to the frozen discriminator and calculating the BCE loss, it prevents the latent vector from degrading into non-geological noise. In this specific implementation, the prior BCE loss is heavily weighted with a multiplier of $\lambda = 10.0$ to guarantee structural integrity during the gradient updates.

Because this latent space optimization often provides a close structural approximation but rarely a perfect voxel-by-voxel match, Pivotal Tuning provides a second phase to resolve this. This second phase uses the optimized noise vector as a static anchor and temporarily unfreezes the generator's internal weights to make microscopic adjustments. This forcefully aligns the localized output with the exact well data, while regularization prevents the collapse of broader geological realism. (A detailed mathematical formulation of this inversion process is provided in Appendix D.1).

## 3.3. Evaluation Metrics

Evaluating the performance of generative models on 3D spatial grids presents significant challenges, as standard metrics like Binary Cross-Entropy (BCE) or direct voxel-by-voxel assessments fail on unconditioned models. Therefore, the evaluation framework relies on metrics that capture complex geological structures and spatial uncertainty across an ensemble of generated samples.

To evaluate how closely the generated realizations resemble the training data's global morphological structures, we employ the Multi-Scale Sliced Wasserstein Distance (MSSWD). Based on the optimal transport principles of the Wasserstein distance (**panaretos2019statistical**), which was highlighted for geological 3D structures by **rongier2025atowards<empty citation>**. MSSWD evaluates structural similarity without requiring rigid pixel alignment. By projecting 3D spatial distributions onto 1D linear slices over multiple downsampled resolution scales and averaging the results, MSSWD efficiently bypasses the computational bottleneck of high-dimensional grid volumes while capturing both local facies boundaries and macro-depositional geometry. It serves as the primary indicator of global architectural consistency and mode collapse during the training process. During training, the model saves 10% of its training data exclusively for evaluation using the MSSWD metric, ensuring these validation samples are never presented to the Discriminator network.

Beyond evaluating global structural realism, quantifying the spatial uncertainty and stochastic diversity across the generated ensemble is critical. To achieve this, voxel-wise Shannon's Entropy (**shannon1948mathematica** is utilized. Let $N$ be the total number of realizations within an ensemble, and let $p_c(z, y, x)$ represent the empirical categorical probability of observing facies class $c \in \{1, \ldots, K\}$ at a specific coordinate $(z, y, x)$, calculated as:

$$p_c(z, y, x) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(\text{Realization}_i(z, y, x) == c) \tag{3.3}$$

where $\mathbb{I}$ is the indicator function. The raw uncertainty at each coordinate is computed via Shannon's entropy field:

$$H_{\text{raw}}(z, y, x) = -\sum_{c=1}^{K} p_c(z, y, x) \log_2 p_c(z, y, x) \tag{3.4}$$

To ensure the metric is bound strictly between 0 and 1, facilitating direct cross-comparison between models with varying numbers of geological classes (e.g., 3-class versus 9-class setups), the field is normalized by the absolute theoretical upper bound $H_{\text{max}} = \log_2(K)$, which occurs under a uniform distribution of pure randomness:

$$H(z, y, x) = \frac{H_{\text{raw}}(z, y, x)}{\log_2(K)} \tag{3.5}$$

By calculating the normalized entropy at each specific spatial coordinate across multiple generated models, this metric serves as a direct validation of the model's stochastic diversity. High entropy values at a given pixel indicate that the generator produces a varied distribution of facies at that location, whereas low entropy exposes areas of rigid uniformity or localized mode collapse.

This voxel-wise evaluation visually maps regions of spatial conflict and agreement, which can be summarized across the entire volume domain to yield a global scalar score $\bar{H} = \frac{1}{V}\sum_{z,y,x} H(z, y, x)$. To evaluate how closely the spatial distribution of uncertainty matches the reference data, the coordinate-wise probability tensors are aggregated and compared using the Jensen-Shannon (JS) divergence (**lin1991divergence**). As a symmetric metric based on Kullback-Leibler divergence (**kullback1951information**), the squared JS divergence measures the coordinate-wise distributional disagreement between the reference baseline and the GAN models (The full mathematical derivations and multi-scale details for these evaluation metrics are expanded in Appendix D.2).

### Data conditioning

While JS divergence defines the overall ensemble spread, after optimization and inversion, each individual GAN model within that ensemble must still strictly honor the physical reality at the wellbore. This physical reality at the wellbore acts as a strict spatial constraint, the localized mismatch of which is calculated using the Macro F1-score. The Macro F1-score calculates the unweighted average of the harmonic mean of precision and recall across all $C$ individual facies classes:

$$F1_{macro} = \frac{1}{C}\sum_{c=1}^{C} \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

Where for each class the precision and recall scores have to calculated:

$$Precision = \frac{TP}{TP + FN}$$

$$Recall = \frac{TP}{TP + FN}$$

where true positive (TP) represents the number of correctly predicted positive pixels, false positive (FP) the predicted positive but actually negative pixels, false negative (FN) the predicted negative but actually positive pixels.

Evaluating every facies class independently provides a highly robust performance metric that heavily penalizes the model if it ignores underrepresented minority classes (i.e. a fair meric when dealing with class imbalance). These underrepresented minority classes would otherwise be completely overshadowed by the more common classes if standard statistical accuracy was deployed to assess the well conditioning.

# 4 Results

The presentation of the results will follow the order of the workflow: First of all the curated datasets will be displayed with the addition of statistical parameters that can be used to evaluate the difference between the samples in the datasets and the synthetic GAN samples. Secondly, the post training results summarize the results of the hyperparameter tuning, the cGAN unconstrained performance to grouped facies classes as well as the cGAN unconstrained performance when presented with all facies classes. Finally, the results of the optimization and inversion results are presented, after the model was constrained to the Delft Geothermal Reservoir well data.

## 4.1. Datasets

In total, 20,000 samples were generated for the three datasets as defined in table 3.1. The generated datasets are visually summarized in figure 4.1: a 2D MDS plot clearly shows a grouping effect per setting, indicating the samples of different settings do in fact display varying higher-dimensional differences. This can be visually confirmed by the two random 3D facies plots per setting, showing that a lower sand body extension as input parameter yields samples with thinner sandbodies and more erratic meandering behavior in samples of this size. The right most plots show the pixel-wise entropy computed over an ensemble of 1000 samples per setting, setting a baseline for the entropy readings we look to achieve from a diverse synthetic dataset produced by the trained GAN model. This threshold configuration can be further evaluated by analyzing table 4.1, which provides an exhaustive overview of the global facies distributions across the ensemble. As can be observed in this overview, the structural fa-

| **Facies** | **Setting 1 (%)** | **Setting 2 (%)** | **Setting 3 (%)** |
|---|---|---|---|
| **1: Channel Lag** | 1.6 | 1.8 | 1.8 |
| **2: Point Bar** | 67.5 | 66.3 | 65.4 |
| **3: Sand Plug** | 1.8 | 1.6 | 1.4 |
| **4: Crevasse Splay Core** | 0.0 | 0.0 | 0.0 |
| **5: Crevasse Channel** | 0.1 | 0.1 | 0.1 |
| **6: Crevasse Splay Delta** | 0.0 | 0.0 | 0.0 |
| **7: Levee** | 5.1 | 6.5 | 7.7 |
| **8: Overbank** | 10.2 | 11.4 | 11.9 |
| **9: Mud Plug** | 13.6 | 12.4 | 11.5 |

Table 4.1: Global categorical facies distribution profiles compared across the three synthetic training datasets for all available classes.

cies, hereafter referred to as classes, exhibit a severe statistical imbalance throughout the data volume. This severe imbalance is a result of dominant classes occupying over 60% of the total grid volume, while corresponding minority classes are entirely absent due to the fundamental boundary limitations of the FLUMY NEXUS parameters. The volumetric presence of classes 4, 5, and 6 is particularly suppressed, yielding an exceptionally low occurrence rate ranging between 0.0% and 0.1%. Altough the classes will be grouped to form three classes instead of nine, the class imbalance will remain with the second class only occuring around 5|5 of the time while the first class occurs approximatly 70% of the time. Such depressed occurrence rates could present an optimization bottleneck as the random occurrence of these volumes throughout the samples could confuse the model into finding an optimized setting that totally disregards these classes. The Discriminator could get stuck on deciding whether a sample is real or fake by judging mostly based on class 1, which leads to the Generator trying to improve the structure of this class, completely disregarding the other classes. Moreover, the high class imbalance in the data has to be accounted for when evaluating the model, especially with regards to the well mismatch evaluation.

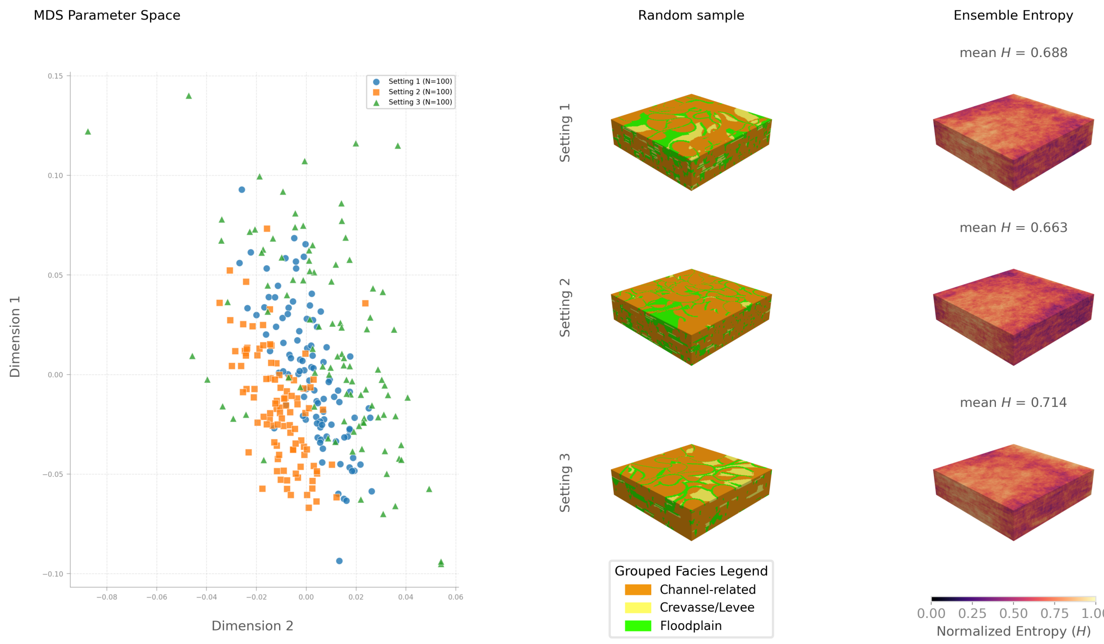



Figure 4.1: MDS plot with two 3D facies samples displayed and a 3D pixel-wise entropy per setting.

## 4.2. Post Training Generative Results

All the models were trained on 2 NVIDIA-A100 GPU's with each 80GB of unified memory. When trained for 25 epochs on 20000 samples, the excecution time took approximately 3-4 hours.

### 4.2.1. Hyperparameter Tuning

Figure 4.2 and table 4.2 provide an overview of the hyperparameter tuning study. All trained model reserved 10% of their training data exclusively for validation: computing the Sliced Wasserstein Distance (SWD) metric for both training and validation datasets. Figure 4.2 includes tracking of the Sliced Wasserstein Distance (SWD) metric across 25 epochs, a randomly generated sample by the trained model and a 3D entropy plot of an ensemble of 1000 GAN generated samples. Complementary, the Jenson-Shannon Divergence (JSD) and Mean SWD which both compare the ensemble of 1000 generated samples per run to a test set of 1000 samples is provided in table 4.2. Based on the metrics from

| Run | Epochs | disc penalty | double conv | double resblock | JSD $\downarrow$ | MS-SWD $\downarrow$ | mean $H_{norm}$ $\uparrow$ |
|---|---|---|---|---|---|---|---|
| 1 | 25 | no | off | off | $0.0718 \pm 0.0157$ | 0.08865 | 0.515 |
| 2 | 25 | no | on | off | $0.1161 \pm 0.0243$ | 0.06842 | 0.430 |
| 3 | 25 | no | on | on | $0.0458 \pm 0.0076$ | 0.08271 | 0.637 |
| 4 | 25 | 32 | on | on | $0.0612 \pm 0.0314$ | 0.09224 | 0.629 |
| **5** | **25** | **16** | **on** | **on** | $\mathbf{0.0155 \pm 0.0067}$ | **0.07986** | **0.704** |
| 6 | 25 | 1 | on | on | $0.0202 \pm 0.0038$ | 0.07690 | 0.614 |
| 7 | 25 | 16 | on | on | $0.1190 \pm 0.0324$ | 0.11924 | 0.393 |
| 8 | 25 | 16 | on | on | $0.0278 \pm 0.0045$ | 0.13389 | 0.623 |

Table 4.2: Overview of hyperparameters and evaluation metrics of each run. Run 5, based on architecture 4 by (**rongier2025atowards**) yields the best JSD and second to best MS-SWD value.

table 4.2 and entropy plots from figure 4.2, the best model was achieved in run 5 with a JSD of 0.0155, MS-SWD of 0.07986 and with the best pixel-wise entropy value out of all the runs (0.79), supported by the 3D entropy plot which shows almost no structure visible in any region. The MS-SWD metric

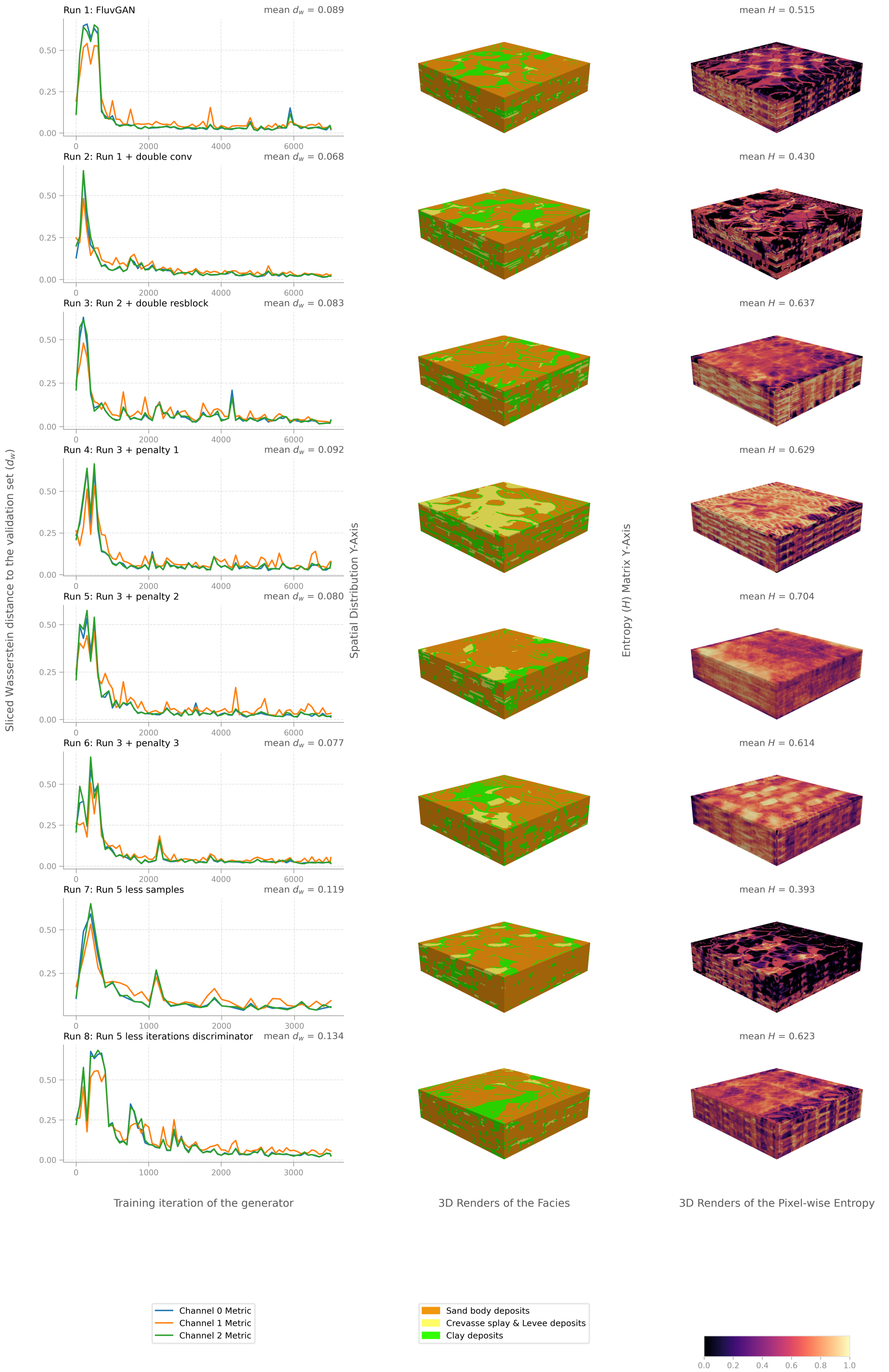



Figure 4.2: Ablation study starting from FluvGan and progressively adding/adjusting parameters to observe their effect on training, geological plausibility and sample diversity. *Run 1:* Standard FluvGAN architecture from (**rongier2025atowards**). *Run 2:* Addition of double convolutional layers. *Run 3:* Addition of double residual blocks (i.e. skip connections) on top of double convolutions. *Run 4:* Addition of a discriminator penalty every 32 iterations. *Run 5:* CHanged discriminator penalty to every 16 iterations. *Run 6:* Changed discriminator penalty to every iteration. *Run 7:* Run 5 hyperparameters but decreased to 10000 training samples. *Run 8:* Run 5 hyperparameters but changed number of itereation of the discriminator to 2.

showed all samples decreasing in Sliced Wasserstein's Distance, indicating a better fit to the training dataset. An interesting observation with regards to the SWD evolution over 25 epochs can be made: All plots show a stabilizing SWD after 2000 training iterations. From a SWD metric standpoint, this indicates that the model has stopped improving its samples compared to the validation dataset.

Moreover, the base FluvGan architecture lead to significant mode collapse looking at figure 4.2. Only after increasing the model depth (double convolutions & double residual blocks) did the model train properly without mode collapse. Moreover, when varying the discriminator R1 Regularization, applying the penalty every 16 iterations yielded the best result. All other option (no penalty, every 32 iterations and every iteration) all resulted in mode collapse. Finally, run 7 and 8 investigated whether decreasing the number of training samples from 20000 to 10000 and only updating the discriminator every 2nd iteration, yielded any improved results. Where the results from run 8 yielded similar results to run 6, the metrics from run 8 yielded a much higher mean JSD of $0.1190 \pm 0.0324$. Additionally, looking at teh entropy plot of run 8 in figure 4.2, The plot shows a significantly higher percentage of regions with low entropy values, indicating that most samples produced by the generator after training have similar pixel-wise values.

### 4.2.2. cGAN Unconstrained Performance

The model with run 5 parameters, of which a complete list with all exact parameters can be found in (add table to the appendix), was run for 100 epochs as well to see whether the results could improve. Altough the the JSD and MS-SWD values did not vary significantly, the geological plausibility of the samples seems to increase significantly, suggested by figure 4.3: when visually, comparing the evolution from 25 to 100 epochs, especially the continuity of the clay deposits, the continuity of this facies seems to have increased (add comparison figure between these two samples! -> 20 3s samples from 25 epochs and 20 samples from 100 epochs.). This is further supported by the blob mean blob size calculation (see table ) As already noted in section **??**, the SWD metric evolution stops improving after around 2000 training iterations, suggesting that the model has stopped improving. However, when assessing the samples from a visual perspective, ... ADD EVOLUTION...

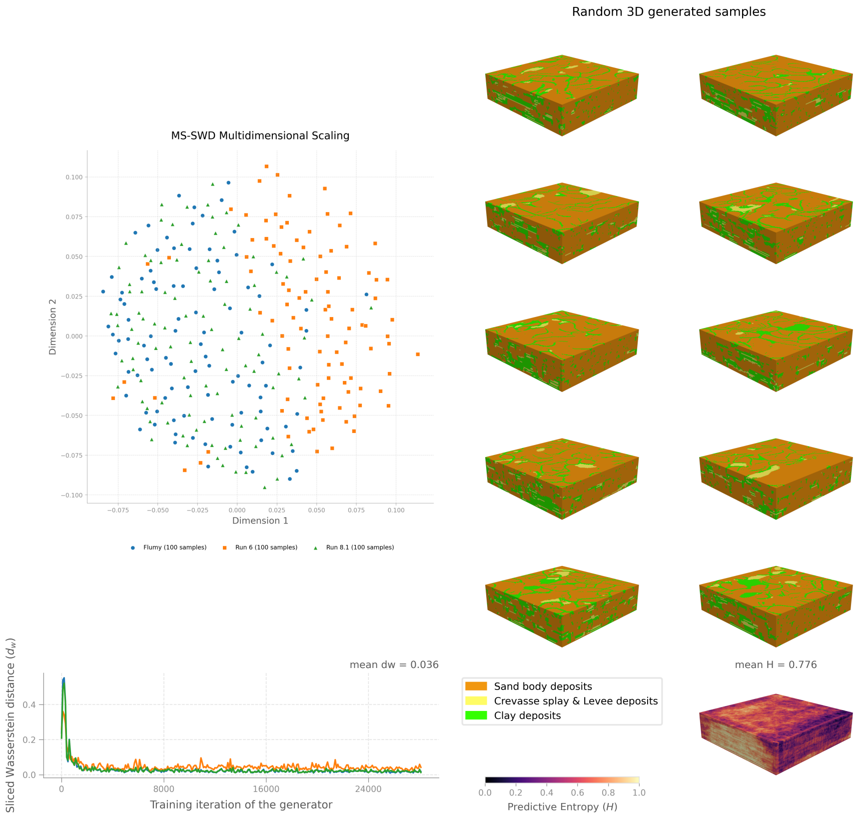



Figure 4.3: Overview of model trained with run 5 settings as described in table 4.2 and figure 4.2 for 100 epochs. The MDS plot shows a 2D representation of a high dimensional structure computed with the Wasserstein distance. For reference have 100 samples from run 6 been added, showing that those samples do not overlap with the training dataset samples, suggesting they lack in full complexity.

### 4.2.3. High-Fidelity 9-Class Generative Performance

Besides only evaluating the model trained on the grouped facies, the training was also conducted on the high fidelity case: testing whether reverting to 9 facies classes lead to significantly different results in terms of geological plausibility and sample diversity. Since the classes are highly unbalanced, this run tests what the capability limits of this model is. The model was trained for 25 epochs on 20000 samples, with the same hyperparameters as run 5. The results are summarized in figure 4.4. The SWD evolution indicates that the models ability to represent class sctuctures, varies significantly per class. For channels 0 to 2 and 6 to 8 the SWD metric decreases immediatly from the start of training, indicating that the model is able to represent these classes well. However, for channels 3 ad 5, the SWD metric no improvement is made, indicating that the model is not able to represent these classes well. Channel 4 decreases over the full amount of iterations with the possibility to decrease even more. Finally, channel 0, channel 2 and channel six, which had a low SWD from the start onwards, display occasional increases in the metric.

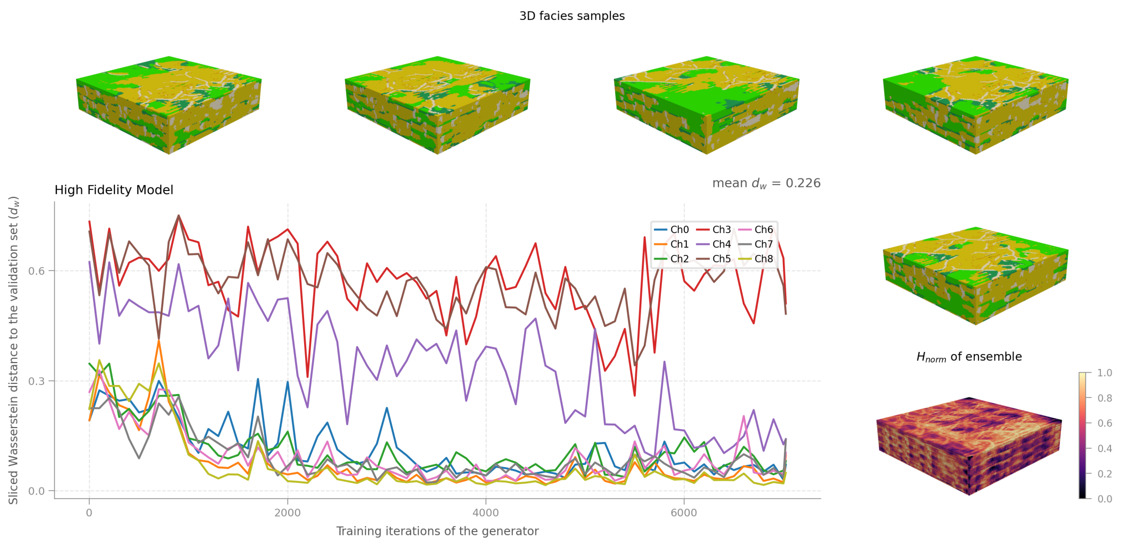



Figure 4.4: Overview of (best) high fidelity model trained over 25 epochs.

From a visual perspective, the samples show signs of being able to recreate geological features. This claim is supported by the Multi-Dimensional Scaling (MDS) plot (ADD FIgure to appendix!) and the corresponding 3D facies plots. However, when evaluating the pixel-wise entropy plot we can observe that the samples show a significant amount of structure, indicating that the model is not able to represent the full complexity of the training dataset. This is further supported by the mean pixel-wise entropy value of ??, which is significantly lower than the mean pixel-wise entropy value of 0.79 for the grouped facies case.

## 4.3. Post-Inversion Results

During optimization and inversion, the tuned model was conditioned to the well data presented in Section 2.1. The evaluation metrics were computed exclusively on the first 32 meters of well data from DEL-GT-01 and DEL-GT-02-S2, yielding a total of 64 spatial conditioning points for the Macro F1-score evaluation. During the initial Latent Space Optimization phase, 100 latent vectors were optimized to fit the well data. These outputs are henceforth referred to as “realizations”, as they represent models explicitly conditioned to real-world subsurface observations.

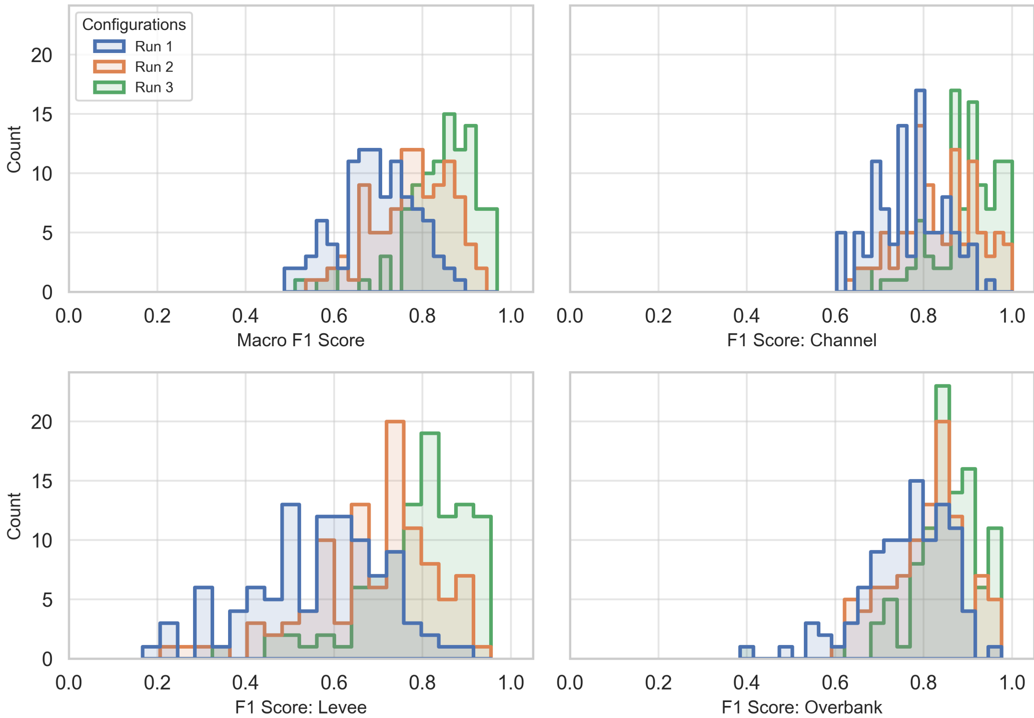



Figure 4.5: Macro F1 and single channel F1 score distributions for three optimization runs with varying threshold values. Run 1: threshold = 1, Run 2: threshold = 2 & Run 3: threshold = 5.

As illustrated in Figure 4.5, the performance of realizations achieving a tight fit with the conditioning points is highly dependent on the optimization threshold ($Th$). The threshold ($Th$) defines a specific spatial radius around the wellbore, measured in voxels, within which the generative model is penalized for mismatching the hard conditioning data. Specifically, voxels located beyond this distance threshold are assigned a weight of zero in the context loss function, ensuring that the inversion process enforces localized geological alignment without over-constraining the broader, unobserved reservoir architecture (See figure D.1). Table 4.3 details the Macro F1-scores, per-class F1 performance, and mean spatial entropy metrics achieved across these varying threshold configurations.

| **Run** | **Th** | **Macro F1**↑ | $F1_{C1}\uparrow$ | $F1_{C2}\uparrow$ | $F1_{C3}\uparrow$ | **Mean** $H_{norm}\uparrow$ |
|---|---|---|---|---|---|---|
| Run 1 | 1 | 0.70 | 0.77 | 0.56 | 0.76 | 0.686 |
| Run 2 | 2 | 0.78 | 0.84 | 0.68 | 0.81 | 0.685 |
| Run 3 | 5 | 0.84 | 0.90 | 0.79 | 0.85 | 0.663 |

Table 4.3: Optimization metrics evaluated across different run configurations. The column ‘Th‘ indicates the specific threshold value defining the spatial region affected by latent optimization. Per-class F1-scores ($F1_{C1}$, $F1_{C2}$, $F1_{C3}$) indicate conditioning accuracy across the individual facies categories at the well locations, accompanied by the global mean normalized spatial entropy ($H_{norm}$) computed over the generated ensemble.

Concurrently, the distribution of realizations achieving high fidelity exhibits a, non-linear response to the threshold parameter. Moving from a tightly constrained threshold of 1 (Run 1) to a wider threshold of 5 (Run 3) yields a substantial increase across all metrics, with the Macro F1-score rising from 0.70 to 0.84. This upward shift is reflected consistently across all three facies classes. For instance, the dominant class ($C1$) scales from an F1-score of 0.77 up to 0.90, while the minority class ($C2$), which typically represents the optimization bottleneck due to severe class imbalance, sees its F1-score jump significantly from 0.56 to 0.79. This systemic improvement suggests that expanding the spatial radius of the context loss mask provides the latent space optimization algorithm with the necessary geometric gradient information to resolve complex multi-scale facies structures, allowing the model to find valid spatial arrangements that honor the hard conditioning points.

Furthermore, analyzing the relationship between conditioning accuracy and ensemble variance highlights a critical spatial trade-off. The main takeaway from the hyperparameter transition is that as the global F1-score increases, the normalized spatial entropy ($H_{norm}$) systematically decreases under wider thresholds. Geologically and mathematically, this behavior occurs because the area under direct deterministic constraint increases radially from the wellbore with higher threshold values. By expanding this optimization zone, a larger volume of the reservoir is forced to adhere strictly to the static well data across the entire ensemble. Consequently, the area characterized by low entropy expands radially outward from the well trajectories, naturally diminishing the overall stochastic diversity and structural variance of the generated realizations in that region (see Appendix **??**).

Importantly, while spatial positioning constraints yield varying F1-scores at the voxel level, the classification capacity of the inverted model remains exceptionally robust. The Receiver Operating Characteristic (ROC) scores are completely perfect across all runs, achieving an Area Under the Curve (AUC $= 1.0$) for all individual facies classes across every optimization configuration (refer to Figure **??**). This flawless AUC metric mathematically confirms that the latent space optimization preserves the pristine probablistic separation and architectural logic of the learned geological prior, ensuring that any minor spatial mismatch captured by the F1 metric is merely a localized boundary shift rather than a breakdown in facies identification.

# 5 Discussion & Recommendations

## 5.1. Interpretations of key results

### 5.1.1. Geological Prior Analysis

The morphological profiles of the curated training datasets show a direct structural control exerted by the underlying process-based parameters. The distinct grouping visible in the 2D MDS space mathematically validates that variations in input parameters translate into unique architectural configurations. Geologically, a lower sand body extension yields significantly thinner sand bodies and a highly erratic meandering behavior. This confirms that the generative model must internalize complex, non-linear spatial frequencies and variable channel tortuosity rather than memorizing a stationary spatial pattern. The pixel-wise entropy computed across the 1,000-sample ensembles establishes a robust high-diversity baseline representing the full stochastic variance of the fluvial-deltaic system, serving as the benchmark for evaluating synthetic diversity.

However, the global categorical facies distribution highlights a severe statistical imbalance that dictates the entire training dynamic. The overwhelming volumetric presence of *Point Bar* deposits, which occupy over 65% of the grid volume across all settings, stands in stark contrast to minority facies like the *Crevasse Splay Core*, *Crevasse Channel*, and *Crevasse Splay Delta*, which are suppressed down to occurrence rates between 0.0% and 0.1% due to fundamental boundary parameters. Even when grouping the nine original facies into three broader classes, the primary class still dominates approximately 70% of the domain while the secondary class appears only 5% of the time. From a machine learning perspective, this creates a critical optimization bottleneck; because a standard GAN treats all voxels with equal weight, the Discriminator is heavily incentivized to act as a shortcut detector, evaluating global sample quality almost entirely on the dominant class. Consequently, the Generator is pushed into a sub-optimal loop that refines large-scale point bars while completely disregarding minority classes, which ultimately complicates both unconstrained generative performance and localized well mismatch evaluations.

### 5.1.2. Architectural Evolution

The hyperparameter ablation study provides vital insights into the structural and regularization configurations required to stabilize deep generative models on high-dimensional 3D spatial data. The severe mode collapse observed in the base FluvGAN architecture (Run 1) proves that a shallow convolutional framework lacks the representational capacity to capture multi-scale fluvial architectures. Overcoming this limitation required a progressive increase in network depth via the integration of double convolutions and double residual blocks, where skip connections successfully preserved high-frequency spatial gradients across the 128x128x32 grid volume. Beyond architecture, the frequency of the discriminator's R1 Regularization penalty proved to be the primary mechanism governing stability, where applying the penalty every 16 iterations (Run 5) hit an optimal sweet spot, yielding the lowest JSD ($0.0155 \pm 0.0067$), a competitive MS-SWD (0.07986), and a well-distributed 3D pixel-wise entropy reaching 0.79. Omitting the penalty or applying it too frequently (Run 6) or infrequently (Run 4) disrupted this delicate equilibrium, driving the model back into mode collapse.

Crucially, the ablation study demonstrates that dataset scale and discriminator update frequencies are non-negotiable parameters for adequate generalization. Halving the training volume from 20,000 to 10,000 samples under identical hyperparameters (Run 7) caused severe performance degradation, resulting in a much higher mean JSD of $0.1190 \pm 0.0324$ and an elevated MS-SWD of 0.11924, mathematically proving that a smaller dataset size is insufficient for the network to resolve complex fluvial configurations. Similarly, altering the discriminator iterations to two updates per generator step (Run 8) led to a significantly higher percentage of low-entropy spatial regions, indicating that the generator began producing highly repetitive, pixel-wise similar outputs. Furthermore, a major disconnect was observed between standard automated distance metrics and true geological plausibility; while the Sliced

Wasserstein Distance (SWD) stabilized and effectively plateaued after approximately 2,000 training iterations, extending the optimal Run 5 model to 100 epochs led to a substantial visual increase in the continuity of clay deposits and mud plugs without significantly changing the JSD or MS-SWD values. This indicates that traditional pixel-distribution metrics are blind to topological connectivity, which takes significantly longer for the network to internalize than global statistical proportions.

### 5.1.3. High-Fidelity and Multi-Class Generative Performance

Reverting from the grouped three-class case to the high-fidelity case with nine distinct facies classes tests the ultimate operational limits of the conditional GAN framework. Managing nine simultaneous channels severely amplifies the structural complexity the network must resolve, especially given the extreme volumetric class imbalance where certain facies dominate over 30% of the volume while minority crevasse associations remain restricted below 0.1%. The primary challenge in this high-fidelity configuration is preventing the network from completely omitting these sparse, thin-bedded deposits, as the mathematical contribution of such a low percentage of voxels to the global discriminator loss is easily overwhelmed by the dominant channels.

Preliminary results from the high-fidelity model trained over 25 epochs indicate that while large-scale structures like main channels and point bars maintain reasonable geological plausibility, the network struggles to accurately preserve the crisp, deterministic boundaries of minor facies. The low occurrence rates act as an optimization bottleneck that confuses the model, occasionally causing it to find a stable loss minimum that completely discards the rare crevasse splay elements. This highlights a clear architectural boundary for unconstrained cGAN models: as the number of categorical facies classes increases alongside severe statistical skewness, standard uniform loss formulations become insufficient, necessitating the future exploration of weighted class loss functions or specialized architectural routing to preserve low-volume geobodies.

### 5.1.4. Post-Inversion Reservoir Realizations

When evaluating the unconstrained cGAN performance, a distinct contrast emerges between macro-scale statistical replication and spatial diversity. While global volume statistics successfully match the geological prior, spatial entropy metrics indicate a significant drop that mathematically confirms a state of ensemble mode collapse where the network repeatedly generates similar spatial configurations rather than exploring the full stochastic variety of the training data. However, the framework excels remarkably during the post-training conditioning phase, demonstrating that generative limitations in unconstrained scenarios do not preclude high-fidelity performance when anchored to hard subsurface data. The two-step inversion process, combining latent space optimization with pivotal tuning of the Generator, successfully forced the stochastically generated volumes to honor localized empirical well logs.

The latent space optimization phase achieved an average macro-F1 score of 0.6, demonstrating a bi-modal distribution where realizations achieved either absolute 100% well matching or fell to approximately $\pm 30\%$ accuracy. Following the subsequent pivotal tuning step, the mean F1 score stabilized further, showcasing the model's capacity to bend mathematically bounded structures to real-world subsurface observations. In practical engineering workflows, the generation of some non-honoring samples is negligible, as users can instantaneously filter out suboptimal realizations and retain the perfectly matched 100% accurate models. When paired with the trained GAN's ability to output a complex 32x128x128 voxel realization in less than one second, this approach provides a highly efficient, real-time application for dynamic uncertainty quantification and reservoir characterization that vastly outperforms traditional, computationally intensive geostatistical simulation methods.

## 5.2. Limitations of the current work

### 5.2.1. Severe Class Imbalance and Optimization Bottlenecks

The process-based training datasets exhibit an extreme categorical disparity that introduces severe algorithmic challenges throughout the machine learning workflow. Dominant sand bodies, specifically Point Bar deposits, occupy over 65% of the global grid volume across all simulated configurations. In stark contrast, vital intermediate and minority architectural elements—such as the Crevasse Splay Core, Crevasse Channel, and Crevasse Splay Delta—are volumetrically suppressed, showing negligible occurrence rates between 0.0% and 0.1% due to fundamental boundary parameters. Even when condensing the high-fidelity case from nine facies down to three broader grouped classes, this systemic imbalance persists, leaving the secondary class at a mere 5% occurrence while the primary class dominates approximately 70% of the grid volume.

From an optimization perspective, this extreme disparity acts as a critical bottleneck. Because the objective function of a standard GAN treats all geocellular voxels with uniform weight, the Discriminator is heavily incentivized to act as a shortcut detector. Instead of evaluating the global geological plausibility of the entire multi-facies system, it can achieve high classification accuracy by judging sample validity solely on the dominant class. Consequently, the Generator is pushed into a suboptimal training loop where it focuses exclusively on refining large-scale point bar structures, completely disregarding minority classes since their total omission yields a negligible penalty to the global loss. This statistical suppression inherently complicates both the unconstrained generative performance and the localized well mismatch evaluations, as the network struggles to accurately represent or optimize structural variations it rarely encounters during the training phase. Resolving this bottleneck requires a transition toward a mathematically rigorous weighted class loss or focal loss formulation to artificially amplify gradients stemming from minority voxels, which remains outside the scope of the current uniform optimization routine.

### 5.2.2. Ensemble Mode Collapse and Uncertainty Underestimation

A fundamental limitation hindering the practical utility of the unconstrained conditional GAN (cGAN) framework is its high susceptibility to ensemble mode collapse. While the model successfully mimics the macro-scale global volume fractions of the geological prior, producing structural proportions that appear nearly identical to the training data, spatial entropy metrics indicate a severe drop. This drop mathematically confirms that the network lacks a rigorous understanding of the true complexity inherent in the synthetic datasets. Instead of exploring the rich stochastic variance and the full architectural diversity of the fluvial-deltaic system, the generator collapses into a narrow subset of stable configurations, repeatedly synthesizing highly similar pixel-wise layouts.

For subsurface engineering and reservoir characterization applications, this structural redundancy severely undermines the efficacy of uncertainty quantification (UQ). By producing an ensemble with artificially deflated spatial variance, the framework severely underestimates the true geological uncertainty present between sparse, localized data points. Consequently, relying on such a collapsed ensemble for downstream hydrodynamic or geothermal fluid flow simulations would yield overly optimistic and uniform connectivity profiles, masking critical operational risks such as early water breakthrough, cold water front short-circuiting, or localized reservoir compartmentalization.

### 5.2.3. Grid Resolution and Scaling Constraints

The fixed spatial discretization of $128 \times 128 \times 32$ voxels imposes significant physical scaling and computational constraints on the generated reservoir models. Under the current configuration, the simulated domain is strictly restricted to a localized volume of $2560$ m $\times$ $2560$ m $\times$ $32$ m. While this grid size is mathematically and computationally sufficient to demonstrate a proof-of-concept inversion workflow—successfully conditioning realizations between two nearby wellbores—it is completely inadequate for capturing full-field architectures.

The localized domain cannot encapsulate the regional-scale depositional sequences, extensive sand-body continuities, and multi-scale heterogeneities characteristic of the entire Delft Sandstone reservoir, introducing severe truncation artifacts along the grid boundaries. Scaling the current framework to full-field dimensions poses massive computational challenges, as a naive expansion of the 3D grid size exponentially amplifies the memory footprint and training time required by the dual-GPU architecture. This resolution ceiling prevents direct integration with macro-scale dynamic reservoir simulators and full-field upscaling workflows, highlighting the necessity for advanced multi-scale or patch-based generative architectures.

## 5.3. Recommendations for future work

Full-Field Upscaling and Dynamic Simulation: Research should expand beyond static geological evaluation by scaling the generated volumes to full-field dimensions using patch-based generation or multi-scale GANs. Furthermore, the 99% conditioned realizations should be subjected to dynamic fluid flow simulations to quantify how the ML-generated architectures impact actual geothermal heat extraction and fluid connectivity compared to traditional geostatistical models.

# 6 Conclusion

The datasets used in this project were generated using the FLUMY geological modelling software. Altough the input parameters varied in the model the datasets show similar characteristics in terms of facies proportions with Point bars being the most abundant facies in the datasets, followed by mud plugs and overbank deposits. The Crevasse play facies achieved a proportions score of 0.0% to 0.1% in all three datasets, indicating that the FLUMY settings used are not fit for recreating these facies.

The FluvGAN model used in this project shows promising results in its ability to recreate geological pausible structures once properly trained on a single prior. This success is highly dependent on the hyperparameter combinations used during training, which showed that a large dataset combined with a moderately occurring discriminator penalty and sufficient network depth are crucial for a run to produce a model not suffering from mode collapse. Additionally, one-hot encoding also proved to a crucial implementation for the model to train properly on categorical data to achieve the correct facies proportions.

## 6.1. Answers to Research Questions

## 6.2. Final Remarks

# A Delft Geothermal Reservoir well data

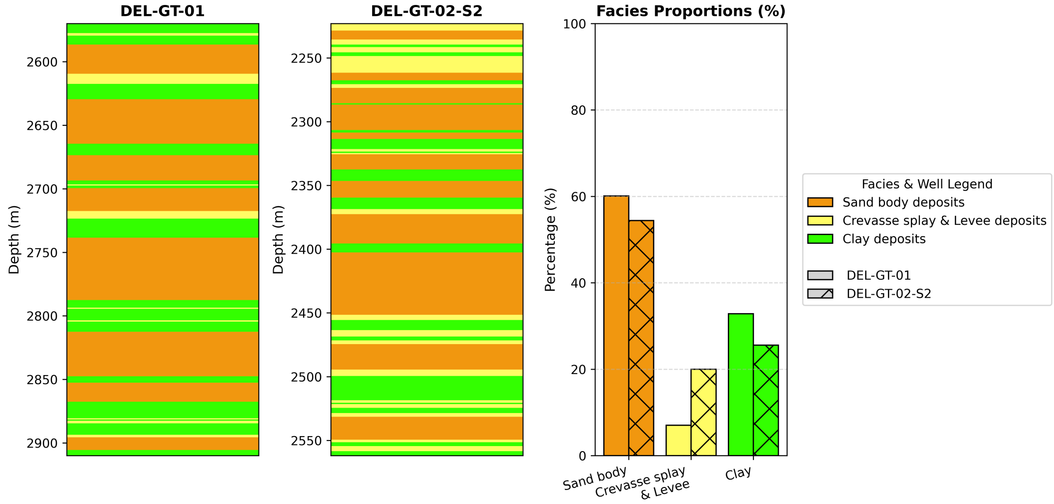
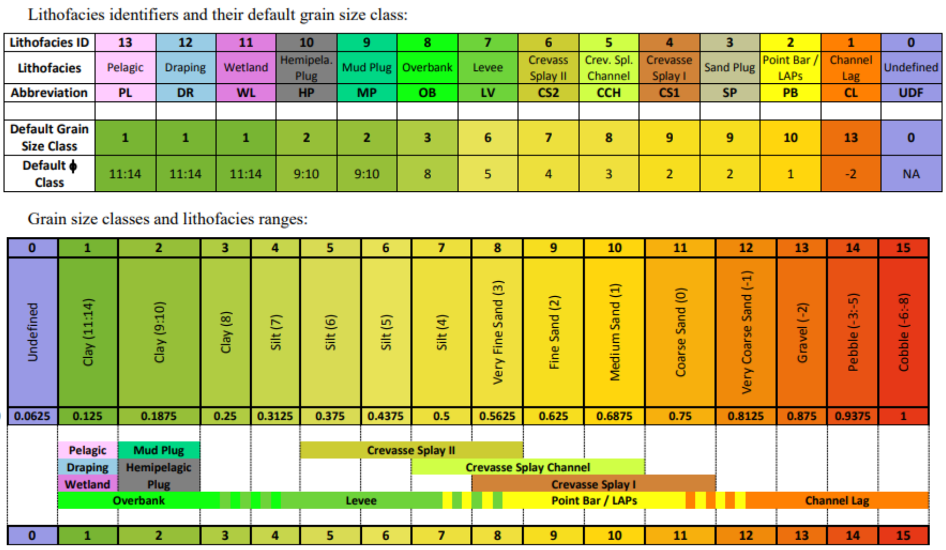



Figure A.1: Overview of Facies associations by (**bruining2023sedimentological**) on the DEL-GT-01 and DEL-GT-02-S2 reservoir section. The Y-axis represents the along hole depth.

# B FLUMY dataset generation

| Value | Facies Key | Color | Description |
|---|---|---|---|
| 0 | Background | | Undefined |
| 1 | Channel Lag | | Coarse active channel fill |
| 2 | Point Bar | | Meandering lateral accretion |
| 3 | Sand Plug | | Fine-grained channel abandonment |
| 4 | Crevasse Splay I | | Proximal high-energy overbank |
| 5 | Crevasse Channel | | Feeder for splay deposits |
| 6 | Crevasse Splay II | | Feeder for splay deposits |
| 7 | Levee | | Bordering channel ridges |
| 8 | Overbank | | Vegetated floodplain silt |
| 9 | Mud plug | | Organic-rich anoxic sediments |
| 10 | Hemipelagic plug | | Fine-grained marine deposits consisting of clay and silt-sized grains |
| 11 | Wetland/Paleosol | | Organic-rich anoxic sediments |
| 12 | Draping | | Fine-grained sediments conforming to topography |
| 13 | Pelagic | | Deep-marine fine-grained suspension settling |

Table B.1: Overview of the 9 facies classes that populated the training dataset. Due to specific setting in the flumy default non expert user settings (see figure 3.1), only 9 facies populated the eventual datasets.


Lithofacies identifiers and their default grain size class:

| Lithofacies ID | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Lithofacies | Pelagic | Draping | Wetland | Hemipela. Plug | Mud Plug | Overbank | Levee | Crevass Splay II | Crev. Spl. Channel | Crevasse Splay I | Sand Plug | Point Bar / LAPs | Channel Lag | Undefined |
| Abbreviation | PL | DR | WL | HP | MP | OB | LV | CS2 | CCH | CS1 | SP | PB | CL | UDF |
| | | | | | | | | | | | | | | |
| Default Grain Size Class | 1 | 1 | 1 | 2 | 2 | 3 | 6 | 7 | 8 | 9 | 9 | 10 | 13 | 0 |
| Default ϕ Class | 11:14 | 11:14 | 11:14 | 9:10 | 9:10 | 8 | 5 | 4 | 3 | 2 | 2 | 1 | -2 | NA |



Figure B.1: Overview of all possible facies defined in the flumy user guide (add citation) with the respecting range of grain sizes.

# C FluvGAN

## C.1. One Hot Encoding Strategy

The one-hot encoding strategy used in this project created $n$ spatial channels based on the number of individual classes passed on to the model. Where traditional one-hot encodiding strategies map the values in a channel to either 1 or 0, referring to the presence or absence of a specific class, the final *Tanh* activation function in the FluvGan model maps values to either -1 or 1. Figure C.1 provides an overview of one hot encoding of a vector. Using one-hot encoding categorical independent data can be correctly fed to the model, avoiding issues where the model think the numbers/values are correlated. The model may start to correlate the the classes based on values, tending to overestimating the classes that do not occur often.

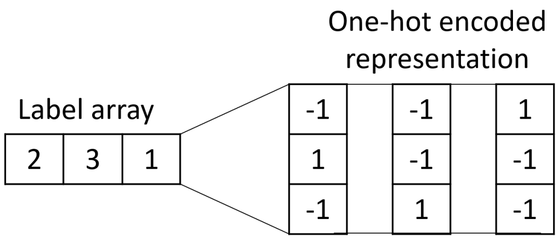



Figure C.1: Simpel one-ho encoding represenation by converting the original data array into a multitude of bipolar vectors. One-hot encoding can be applied to any type of categorical data.

## C.2. Fluvgan Architecture

| **Layer / Block Type** | **Output Shape** | **Param #** |
|---|---|---|
| ConvBlockTranspose3d (Scale 1) | [1, 1024, 4, 4, 4] | 6,553,600 |
| DeepResBlockUpscale3d (Scale 2) | [1, 1024, 8, 8, 8] | 4,066,816 |
| DeepResBlock3d (Scale 2) | [1, 1024, 8, 8, 8] | 4,066,816 |
| DeepResBlockUpscale3d (Scale 3) | [1, 512, 16, 16, 16] | 3,935,744 |
| DeepResBlock3d (Scale 3) | [1, 512, 16, 16, 16] | 1,017,600 |
| DeepResBlockUpscale3d (Scale 4) | [1, 256, 32, 32, 32] | 984,832 |
| DeepResBlock3d (Scale 4) | [1, 256, 32, 32, 32] | 254,848 |
| DeepResBlockUpscale3d (Scale 5) | [1, 128, 32, 64, 64] | 99,200 |
| DeepResBlock3d (Scale 5) | [1, 128, 32, 64, 64] | 27,072 |
| DeepResBlockUpscale3d (Scale 6) | [1, 64, 32, 128, 128] | 25,024 |
| DeepResBlock3d (Scale 6) | [1, 64, 32, 128, 128] | 6,880 |
| BatchNorm3d + LeakyReLU | [1, 64, 32, 128, 128] | 128 |
| ConvBlock3d (Output) | [1, 9, 32, 128, 128] | 15,552 |
| **Total Trainable Parameters** | **21,054,112** | |

Table C.1: Detailed architectural specifications and layer parameters for the DeepGenerator3d network.

| **Layer / Block Type** | **Output Shape** | **Param #** |
|---|---|---|
| ConvBlock3d (Input) | [1, 64, 32, 128, 128] | 15,552 |
| DeepResBlockDownscale3d (Scale 2) | [1, 128, 16, 64, 64] | 20,992 |
| DeepResBlock3d (Scale 2) | [1, 128, 16, 64, 64] | 63,488 |
| DeepResBlockDownscale3d (Scale 3) | [1, 256, 8, 32, 32] | 83,968 |
| DeepResBlock3d (Scale 3) | [1, 256, 8, 32, 32] | 253,952 |
| DeepResBlockDownscale3d (Scale 4) | [1, 512, 4, 16, 16] | 335,872 |
| DeepResBlock3d (Scale 4) | [1, 512, 4, 16, 16] | 1,015,808 |
| DeepResBlockDownscale3d (Scale 5) | [1, 1024, 4, 8, 8] | 753,664 |
| DeepResBlock3d (Scale 5) | [1, 1024, 4, 8, 8] | 1,703,936 |
| DeepResBlockDownscale3d (Scale 6) | [1, 1024, 4, 4, 4] | 1,703,936 |
| DeepResBlock3d (Scale 6) | [1, 1024, 4, 4, 4] | 1,703,936 |
| LeakyReLU | [1, 1024, 4, 4, 4] | 0 |
| LinearBlock (Output Prediction) | [1, 1] | 1,025 |
| **Total Trainable Parameters** | **7,656,129** | |

Table C.2: Detailed architectural specifications and layer parameters for the DeepDiscriminator3d network.

# D Mathematical Formulations for Optimization and Evaluation

## D.1. Optimization & Inversion Framework

The two-step inversion process for conditioning the 3-facies generative models is mathematically defined by consecutive optimization phases: Latent Space Optimization followed by Pivotal Tuning.

### D.1.1. Latent Space Optimization

During the initial Latent Space Optimization, the internal weights of the generator $G$ are frozen. Gradient descent is utilized to iteratively search the latent space $\mathcal{Z}$ for an optimal latent noise vector $\mathbf{z}^*$ that minimizes the loss between the generated facies and the true geological observations.

Rather than evaluating the penalty exclusively at the exact well coordinates, a spatially weighted context mask is applied to enforce localized geological alignment without over-constraining the broader, unobserved reservoir architecture. A spatial influence radius is established around each well point, governed by a distance threshold ($\sigma$). This threshold acts as the standard deviation for a continuous Gaussian decay function, dictating the geostatistical "reach" of the well data.

Because fluvial-deltaic depositional systems are highly anisotropic—characterized by extensive horizontal continuity and limited vertical thickness—an anisotropic distance metric $d_A(v)$ is utilized, penalizing vertical deviations significantly more than horizontal ones. For any voxel $v$ in the 3D grid $\mathcal{V}$, the spatial weight mask $M(v)$ is defined as:

$$M(v) = \begin{cases} \exp\left(-\frac{d_A(v)^2}{2\sigma^2}\right), & \text{if } d_A(v) \leq 3\sigma \\ 0, & \text{if } d_A(v) > 3\sigma \end{cases} \tag{D.1}$$

While the threshold ($\sigma$) controls the smooth, continuous spread of the penalty, calculating gradients for distant voxels is computationally inefficient. Therefore, the mask is explicitly truncated to zero at a hard computational cutoff of $3\sigma$, where the gradient contribution becomes mathematically negligible.

The objective function is thus defined as a spatially masked Context Loss, replacing the point-wise evaluation. It is evaluated over the grid volume $\mathcal{V}$ and the $C$ total facies classes:

$$\mathbf{z}^* = \arg\min_{\mathbf{z}} \left[ \sum_{v \in \mathcal{V}} \sum_{i=1}^{C} M(v) \cdot \mathcal{L}_{CE}(p_{v,i}, y_{v,i}) \right] \tag{D.2}$$

where $\mathcal{L}_{CE}$ is the Categorical Cross-Entropy. Specifically, $y_{v,i}$ is the binary truth value (0 or 1) indicating whether class $i$ is the correct broadcasted facies for voxel $v$ (based on the nearest well data point), and $p_{v,i}$ is the generator's predicted probability for that class.

**Computational Scaling:** Latent optimization is highly efficient. The computational cost scales linearly, $\mathcal{O}(N_w \cdot C)$, with the number of conditioning points $N_w$ and facies classes $C$. Because the network weights are frozen, the gradient is only computed with respect to the latent vector $\mathbf{z}$ (typically of dimension 512 or 1024), making this phase fast and highly scalable to dense well configurations.

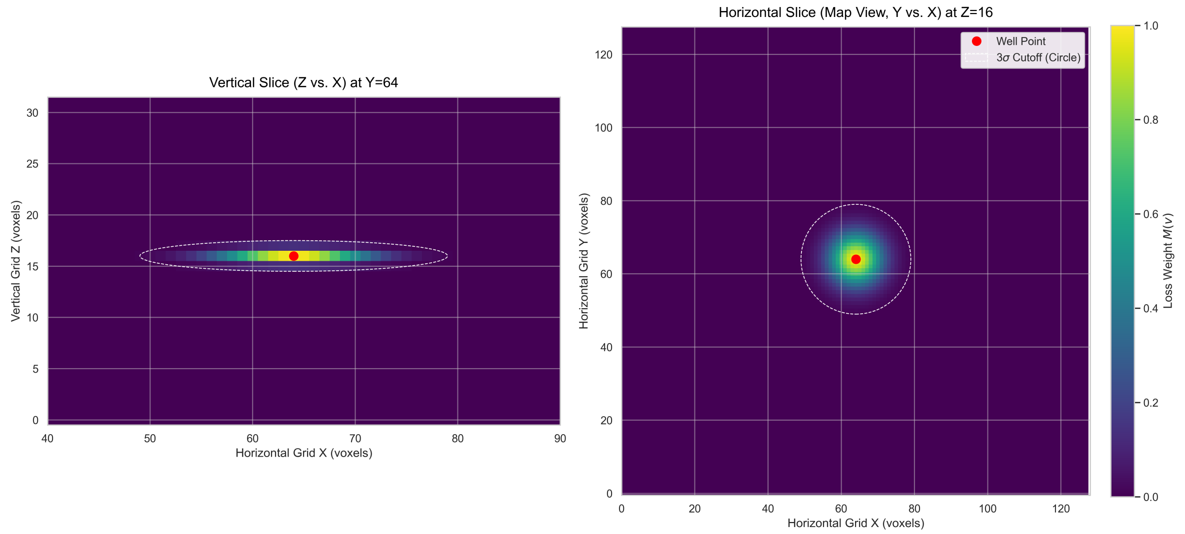



Figure D.1: Orthogonal cross-sections illustrating the 3D anisotropic Gaussian weight mask $M(v)$ applied during Latent Space Optimization. The anisotropy factors are 10,1,1. The distance threshold ($\sigma$) controls the smooth spatial decay of the penalty away from the well point (red dot). Due to the anisotropic scaling factor, the mask forms a horizontal lens in the vertical cross-section (left), respecting stratigraphic deposition, while maintaining a perfectly radial spread in the map view (right). The gradient is safely truncated at a $3\sigma$ limit (dashed line) to preserve computational efficiency.

### D.1.2. Pivotal Tuning

To overcome the inherent boundaries of the learned latent space, Pivotal Tuning (**roich2022pivotal**) is applied. Using the previously optimized $\mathbf{z}^*$ as a static anchor, the generator's internal weights $\theta$ are unfrozen and optimized:

$$\theta^* = \arg\min_{\theta} \mathcal{L}_{CE}(G_\theta(\mathbf{z}^*)_{\text{well}}, \mathbf{y}_{\text{well}}) + \lambda||\theta - \theta_0||_2 \tag{D.3}$$

This formulation pushes the localized outputs to perfectly match the hard conditioning data. The critical addition here is the $L_2$ regularization term $\lambda||\theta - \theta_0||_2$, where $\theta_0$ represents the original pre-trained weights and $\lambda$ is a penalty coefficient. This term forces the modified weights to stay as close as possible to their original state, preserving the global geological realism of the model and preventing catastrophic forgetting.

**Computational Scaling:** Unlike latent optimization, Pivotal Tuning is computationally intensive. The optimization scales with the total number of trainable parameters in the generator, $|\theta|$, which often numbers in the tens of millions for 3D generative networks, $\mathcal{O}(|\theta| \cdot N_w)$. It requires backpropagation through the entire network architecture for every iteration, scaling linearly with the size of the 3D grid output during the forward pass.

## D.2. Generative Evaluation Metrics

The evaluation framework relies on advanced statistical divergence and optimal transport metrics to overcome the limitations of standard voxel-by-voxel assessments in unconditioned 3D modeling.

### D.2.1. Wasserstein Distance and Mean Sliced Wasserstein Distance

The conceptual foundation for structural evaluation is the Wasserstein distance, which quantifies the optimal transport cost required to reconfigure the generated probability distribution $P_g$ into the reference real data distribution $P_r$ (**panaretos2019statistical**; **rongier2025atowards**). It is defined as:

$$\mathcal{W}(P_r, P_g) = \inf_{\gamma \sim \Pi(P_r, P_g)} \mathbb{E}_{(x,y)\sim\gamma}[||x - y||] \tag{D.4}$$

where $\Pi(P_r, P_g)$ denotes the set of all joint probability distributions $\gamma(x, y)$ whose marginals are $P_r$ and $P_g$ respectively.

Because computing Equation **??** over high-dimensional 3D grids suffers from the curse of dimensionality, we employ the Mean Sliced Wasserstein Distance (MSWD). MSWD projects the complex 3D distributions onto 1D lines where the optimal transport problem has a trivial, closed-form solution (sorting). Mathematically, it is defined by integrating the 1D Wasserstein distances over the unit hypersphere $\mathbb{S}^{d-1}$:

$$\text{MSWD}(P_r, P_g) = \int_{\mathbb{S}^{d-1}} \mathcal{W}_{1D}(\mathcal{R}_\omega \# P_r, \mathcal{R}_\omega \# P_g) d\omega \tag{D.5}$$

where $\mathcal{R}_\omega \# P$ denotes the projection (Radon transform) of distribution $P$ onto a 1D line defined by the direction vector $\omega$. In practice, this integral is approximated using a Monte Carlo approach by averaging over a large number of random projection angles $L$.

**Computational Scaling:** Computing the exact Wasserstein distance scales at roughly $\mathcal{O}(V^3 \log V)$, where $V$ is the total number of voxels in the 3D grid. This is intractable for standard geomodels. MSWD circumvents this by scaling at $\mathcal{O}(L \cdot V \log V)$, where $L$ is the number of projection slices. Since sorting 1D arrays is exceptionally fast, MSWD allows for robust structural comparison of massive 3D grids in a fraction of the time.

### D.2.2. Uncertainty and Spatial Divergence Metrics

To assess spatial uncertainty and validate generation diversity across an ensemble, Normalized Shannon's Entropy (**shannon1948mathematical**) is calculated independently for every spatial coordinate (voxel) $\mathbf{v} \in \mathbb{R}^3$. It scales the standard entropy equation by the number of geological classes ($k$) to bound the metric between 0 and 1:

$$H_{norm}(\mathbf{v}) = -\frac{1}{\log_2(k)} \sum_{i=1}^{k} P(F_i|\mathbf{v}) \log_2 P(F_i|\mathbf{v}) \tag{D.6}$$

Here, $P(F_i|\mathbf{v})$ is the empirical probability of generating facies $F_i$ at the specific voxel $\mathbf{v}$ across the entire ensemble of realizations. An $H_{norm}(\mathbf{v})$ of $0$ implies total certainty (uniformity or potential mode collapse at that pixel), while $1$ implies total uncertainty (maximum diversity across all $k$ facies).

**Computational Scaling:** Entropy calculations are highly efficient parallel operations. The scaling is purely linear, $\mathcal{O}(V \cdot k)$, where $V$ is the total number of voxels, making it perfectly suited for massive 3D grids regardless of ensemble size.

To quantify the distributional disagreement between the true reference models ($P$) and the generated models ($Q$), we calculate the pixel-wise Kullback-Leibler (KL) divergence (**kullback1951information**):

$$D_{KL}(P_\mathbf{v} \parallel Q_\mathbf{v}) = \sum_{i=1}^{k} P(F_i|\mathbf{v}) \log_2 \left( \frac{P(F_i|\mathbf{v})}{Q(F_i|\mathbf{v})} \right) \tag{D.7}$$

Because KL divergence is asymmetric and unbounded, it is transformed into the Jensen-Shannon (JS) divergence (**lin1991divergence**) to provide a bounded, symmetric measure of stochastic spread at each voxel:

$$D_{JS}(\mathbf{v}) = \frac{1}{2} D_{KL}(P_\mathbf{v} \parallel M_\mathbf{v}) + \frac{1}{2} D_{KL}(Q_\mathbf{v} \parallel M_\mathbf{v}) \tag{D.8}$$

where $M_\mathbf{v} = \frac{1}{2}(P_\mathbf{v} + Q_\mathbf{v})$ is the mixture distribution at voxel $\mathbf{v}$.

Finally, to aggregate this 3D grid of spatial conflict into a single definitive scalar for model evaluation, the mean JS divergence is computed across the entire spatial volume $\mathcal{V}$:

$$\overline{D}_{JS} = \frac{1}{|\mathcal{V}|} \sum_{\mathbf{v} \in \mathcal{V}} D_{JS}(\mathbf{v}) \tag{D.9}$$

**Computational Scaling:** Similar to entropy, JS Divergence operates strictly on the pre-computed probability mass functions of the respective ensembles. Therefore, the scalar aggregation scales linearly with the number of voxels evaluated, $\mathcal{O}(V)$. It remains computationally trivial even for massive, high-resolution spatial domains.

To aggregate this spatial conflict into a single quantitative scalar comparing two datasets, we utilize the Jensen-Shannon (JS) divergence (**lin1991divergence**). It measures the symmetric stochastic spread between the true reference distributions ($P$) and the generated distributions ($Q$):

$$D_{JS}(P \parallel Q) = \frac{1}{2} D_{KL}(P \parallel M) + \frac{1}{2} D_{KL}(Q \parallel M) \tag{D.10}$$

where $M = \frac{1}{2}(P + Q)$ is a mixture distribution, and $D_{KL}$ is the standard Kullback-Leibler divergence (**kullback1951information**), explicitly formulated as:

$$D_{KL}(P \parallel Q) = \sum_{x} P(x) \log \left( \frac{P(x)}{Q(x)} \right) \tag{D.11}$$

Unlike $D_{KL}$, which is asymmetric and can approach infinity if $Q(x)$ is zero, $D_{JS}$ is strictly symmetric ($D_{JS}(P \parallel Q) = D_{JS}(Q \parallel P)$) and bounded between $[0, 1]$ when calculated with base-2 logarithms, making it a robust final scalar for spatial fidelity.

**Computational Scaling:** Similar to entropy, JS Divergence operates strictly on the pre-computed probability mass functions of the respective ensembles. Therefore, it scales linearly with the number of spatial bins/voxels evaluated, $\mathcal{O}(V)$. It remains computationally trivial even for massive, high-resolution spatial domains.